



Resident Space Object Characterization Using Polarized Light Curves

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Light curves, or the time-history of photometric brightness, have previously been demonstrated to allow for estimation of a space object's attitude, shape, and surface parameters. However, these methods are subject to many limitations, including the need for accurate initial estimates to initialize filters, and the inability to regularly estimate attitude and surface parameters simultaneously due to information dilution, or the increase in uncertainty of each state when more states are simultaneously estimated. Monochromatic, unpolarized light curves can also only allow for surface reflectance parameter determination, and not determination of the materials themselves. This work develops a polarized bidirectional reflectance distribution function, and demonstrates that polarized measurements increase the uniqueness of light-curve attitude solutions to allow larger tolerance to initial error while also providing surface material estimation. Polarization of reflected light can also be measured more accurately than the intensity of reflected light through the atmosphere, thereby increasing the attitude observability from the light curves. Observability of attitude and surface materials is assessed. An unscented multiple-model adaptive estimation filter is implemented to estimate attitude, angular rate, surface material, surface roughness, and size of the space object.

I. Introduction

SPACE situational awareness (SSA) refers to “the knowledge and characterization of resident space objects (RSOs) and their operational environment to support safe, stable, and sustainable space activities” [1]. As the number of RSOs in orbit grows, it becomes increasingly important to be able to accurately propagate the motion of each object forward in time in order to predict whether two objects are at risk of colliding. A major limiting factor in accurately propagating an RSO's orbit is the ability to account for nonconservative perturbations, notably drag and solar radiation pressure (SRP). The accelerations due to drag and SRP both depend on the cross-sectional area of the object projected along the velocity vector and sun vector, respectively. Acceleration due to SRP also depends on the reflectivity of the surface materials. Due to the dependence of these perturbations on the cross-sectional area and surface reflectivity, there is a coupling between an RSO's orbit and its attitude, shape, mass, and surface materials. However, most orbit propagation for catalog maintenance assumes that the object is a “cannonball,” with a constant area-to-mass ratio. Note that the area-to-mass ratio for drag and SRP is different, as the cross-sectional area is projected along a different vector. However, in practice, often only one area-to-mass ratio or ballistic coefficient is estimated; typically drag for low Earth orbit (LEO) and SRP for high-altitude orbits such as geosynchronous orbit (GEO). This simplifying assumption of constant area-to-mass ratio is a large source of error in the orbit propagation. Compared to higher-fidelity models, errors of 1 km can accumulate over approximately 4 days for a typical spacecraft in GEO. For high area-to-mass ratio debris objects in GEO, errors of several hundred kilometers per day are possible [2].

To overcome these shortcomings in the cannonball model, knowledge of the RSO's shape and size, mass, attitude, and surface materials is necessary. However, while astrometric data alone can provide for approximation of a ballistic coefficient or area-to-mass ratio, it does not provide sufficient information about other more detailed attributes. Estimation and characterization of these attributes requires information from a different source. If resolved imagery is available, then a shape attribute may be observed directly from the image. For objects in LEO, radar cross sections can provide additional information on an object's size and shape. However, for objects in higher-altitude orbits such as GEO, radar cross sectioning is ineffective due to the distance involved. One approach to estimating these attributes is through use of the RSO's photometric signature, or light curve. A light curve is the time-history of photometric brightness.

Prior work has established the viability of using the light curve to estimate attributes of an RSO. In [3] determination of the coupled albedo-area of an object as a function of the surface normal is demonstrated, but this requires knowledge of the object's attitude. In [4] estimation of both attitude and shape parameters from the light curve is demonstrated; however, this method is limited to convex shapes, which is a poor assumption for many RSOs. Model-based methods, which assume a shape model or a bank of potential shape models, have been demonstrated to allow for characterization of many of an RSO's attributes. In [5] it is shown that static photometric observations of an RSO contribute information about its attitude, even though the attitude itself is not fully observable from a single observation location. With multiple facets and observation locations, the attitude is found to become fully observable. This work also demonstrated the observability of surface reflectance parameters, but is limited to the properties of the reflectance models and not the materials themselves. In [6] the dynamic observability of attitude, angular velocity, shape, and surface parameters is demonstrated. For the dynamic case, the photometric brightness over time is used, and it is seen that the attitude, shape, and surface parameters are observable from measurements from a single location. In [7] attitude estimation using light curves is demonstrated, assuming that shape and surface reflectance properties are well-known. In [8], simultaneous orbit, attitude, and surface parameter estimation from angles (astrometric) and light curve (photometric) data is demonstrated. However, this work showed that it is often difficult to simultaneously estimate surface parameters with attitude, likely due to information dilution. “Information dilution” refers to the increase in uncertainty of state

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estimates as additional states are simultaneously estimated given a fixed measurement set. In [9] simultaneous shape, surface parameter, and attitude profile information using a multiple-hypothesis method is demonstrated. However, this method assumes several common attitude states, which are included within the models and are not estimated as filter states, so this method is ineffective for arbitrary attitudes. In [10] shape estimation in conjunction with attitude estimation using multiple-model adaptive estimation is demonstrated. This method is effective for real RSOs when compared to the methods of [3,4] as it allows for nonconvex, complex shapes and does not require parameterization of the shapes. In [11] attitude estimation with shape-model uncertainty is demonstrated by treating the effects of shape mismodeling as a measurement bias.

One issue with attitude estimation from light curves is that the filters are not very robust to initial attitude errors. For example, the filter presented in [7] can only tolerate initial errors of a few degrees when tested with a representative shape model. This is a severe limit on the usefulness of these approaches, as it is unlikely that attitude would be initially known to this degree of accuracy. There are several factors that may contribute to this issue. The first is a lack of sufficient information being contributed by the light curve. The light curve is only one measurement, which can have a highly nonlinear coupling to the spacecraft attitude. Therefore, it can be difficult to correctly determine a three-dimensional attitude from noisy light curve measurements. Also, nonconvex shape models may produce many local minima in an associated light-curve residual. It has been found that there often exist many attitudes that locally minimize the light curve residual. If the filter initially starts driving toward one of these local minima that is not the global minimum, it is unlikely that the filter will ever converge properly. Therefore, it is desirable to investigate additional sources of information that may make this problem better conditioned.

To date, very little work has been performed to assess the usefulness of polarized light curves in estimating space object attributes. In [12] the polarized reflection from a spacecraft using a microfacet model is modeled. Specifically, this approach models the specular reflection off a surface, as well as subsurface scattering. The model is created as a collection of microfacets, and the polarized reflection off each facet is computed in parallel using a graphical processing unit.

In this work, the usefulness of polarized light curves in estimating attributes of an RSO is assessed. The polarization of diffusely reflected light is shown to depend on the material from which the light is reflecting, as well as the angle at which the reflecting facet is being viewed. Therefore, if the positions of the object and observer are known, the polarization contains additional information about the RSO's attitude. Additionally, while the polarization of the specular reflection is found not to depend on the object's attitude, it still has dependence on the material properties. These dependencies have the potential to allow for better attribute estimation performance compared to the unpolarized light curves.

The outline of this paper is as follows. First, reflection models to model polarized light curves will be developed based on existing bidirectional reflectance distribution functions (BRDFs), but with the addition of polarization of the specular and diffuse components. The assumptions and limitations of this model will be addressed, as will the material properties necessary to simulate polarized light curves. Special cases will also be considered, such as the case of compound materials, where a transparent or semitransparent material is covering another material, such as in spacecraft solar panels or multilayer insulation. The use of polarized light curves for attitude and surface material determination will then be analyzed. Sensitivity to model parameters and modeling errors will be assessed. This will include the performance in the presence of depolarization, which can be due to unmodeled effects in the reflecting material's surface. Methods to quantify and determine the attitude observability from polarized light curves will be developed, and this observability will be compared to unpolarized light curves to give insight into the benefit of utilizing polarization data. The observability of surface parameters will also be evaluated. Attitude determination performance, including filter convergence and the ability to handle initial errors, will also be assessed.

II. Light Curve Models

A. Unpolarized Light Curve Model

The following BRDF model is used as the basis for further sensor modeling and polarized model development. The model described in this section is that of [13] with the modification of [14]. This model is used in [5,8,10]. The apparent magnitude m_{app} is computed using the BRDF model and a shape model. The reflection geometry for a single facet is shown in Fig. 1. Each facet has a set of three basis vectors ($\mathbf{u}_n^B$, $\mathbf{u}_u^B$, $\mathbf{u}_v^B$), expressed in the body frame B . The unit vector $\mathbf{u}_n^B$ points in the direction of the outward normal of the facet, and the vectors $\mathbf{u}_u^B$ and $\mathbf{u}_v^B$ are in the plane of the facet. In this analysis, the object is assumed to be a rigid body so that these unit vectors do not change since they are expressed in the body frame. The vector $\mathbf{u}_h^I$ is the normalized half vector between $\mathbf{u}_{sun}^I$, the unit vector to the sun, and $\mathbf{u}_{obs}^I$, the unit vector to the observer, expressed in the inertial frame I . The observation vector is usually given in body coordinates with $\mathbf{u}_{obs}^B = A\mathbf{u}_{obs}^I$, where A is the inertial-to-body attitude matrix. The BRDF at any point on the surface is a function of two directions, the direction from which the light source originates and the direction from which the scattered light leaves the surface. The BRDF is decomposed into a specular component and a diffuse component. The two terms sum to give the total BRDF for the i th facet:

$$\rho_{total,i} = \rho_{spec,i} + \rho_{diff,i} \quad (1)$$

The diffuse component represents light that is scattered equally in all directions and the specular component represents light that is concentrated about some direction. The diffuse component is a Lambertian-like term, but is non-Lambertian in that it enforces energy conservation in the presence of specular reflection. Pure Lambertian reflection has the same apparent brightness regardless of an observer's angle of view, and does not take into account the presence of specular reflections. Ashikmin and Shirley [13] develop a model for continuous arbitrary surfaces but simplify it for flat surfaces. This simplified model is employed for the simulations in this work and all further model developments as the shape models are considered to consist of a finite number of flat facets. Therefore the total observed brightness of an object becomes the sum of the contribution from each facet.

Under the flat facet assumption, the specular term of the BRDF becomes

$$\rho_{spec,i} = \frac{\sqrt{(n_u + 1)(n_v + 1)}}{8\pi} \times \frac{(\mathbf{u}_{n,i}^I \cdot \mathbf{u}_h^I)^z}{\mathbf{u}_{n,i}^I \cdot \mathbf{u}_{sun}^I + \mathbf{u}_{n,i}^I \cdot \mathbf{u}_{obs}^I - (\mathbf{u}_{n,i}^I \cdot \mathbf{u}_{sun}^I)(\mathbf{u}_{n,i}^I \cdot \mathbf{u}_{obs}^I)} F_{reflect,i} \quad (2a)$$

$$z = \left\{ \frac{n_u (\mathbf{u}_h^I \cdot \mathbf{u}_{u,i}^I)^2 + n_v (\mathbf{u}_h^I \cdot \mathbf{u}_{v,i}^I)^2}{(1 - [\mathbf{u}_{n,i}^I \cdot \mathbf{u}_h^I]^2)} \right\} \quad (2b)$$

where n_u and n_v define distribution of the specular lobe in the u and v directions, respectively, and $F_{reflect,i}$ is the Fresnel reflectance. The Fresnel reflectance, for nonpolarized observations, is defined as [15]

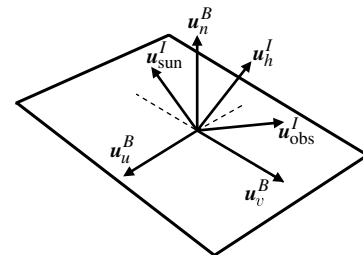


Fig. 1 Facet reflection geometry.

$$F_{\text{reflect},i} = \frac{1}{2} \frac{(a-c)^2 + b^2}{(a+c)^2 + b^2} \left[\frac{(a+c-1/c)^2 + b^2}{(a-c+1/c)^2 + b^2} + 1 \right] \quad (3)$$

with

$$a^2 = \frac{1}{2} \left(\sqrt{(n^2 - k^2 + c^2 - 1)^2 + 4n^2k^2} + n^2 - k^2 + c^2 - 1 \right) \quad (4a)$$

$$b^2 = \frac{1}{2} \left(\sqrt{(n^2 - k^2 + c^2 - 1)^2 + 4n^2k^2} - n^2 + k^2 - c^2 + 1 \right) \quad (4b)$$

$$c = \mathbf{u}_h^I \cdot \mathbf{u}_{\text{sun}}^I = \mathbf{u}_h^I \cdot \mathbf{u}_{\text{obs}}^I \quad (4c)$$

where n is the material's index of refraction and k is its extinction coefficient. However, if multispectral light curves are to be rendered, then n and k depend on the wavelength, e.g., n_λ and k_λ . However, n_λ and k_λ are often unknown across the spectrum, and only mean values $\bar{n}$ and $\bar{k}$ are available. In contrast, the spectral reflectance at normal incidence $R_{\text{spec},\lambda}$ has been measured across the visible spectrum for thousands of materials. Thus, Schlick [15] makes the following approximation:

$$F_{\text{reflect},i} = R_{\text{spec},i} + (1 - R_{\text{spec},i})(1 - c)^5 \quad (5)$$

This approximation is employed in the light curve model of [13]. If the surface was perfectly smooth, all of the Fresnel reflectance would be directed along the reflectance angle, expressed in body coordinates as

$$\mathbf{u}_{\text{reflect},i}^B = \mathbf{u}_{\text{sun}}^B - 2(\mathbf{u}_{\text{sun}}^B \cdot \mathbf{u}_{n,i}^B) \mathbf{u}_{n,i}^B \quad (6)$$

where $\mathbf{u}_{\text{sun}}^B$ is the sun vector in body coordinates. However, real materials have some level of surface roughness, which results in many small microfacets that have surface normals oriented in different directions. Therefore, the specular reflectance can be thought of as a cone distributed around the specular reflectance direction given in Eq. (6), with the distribution (in the local u and v directions) defined by the terms n_u and n_v . If the material is isotropic, then n_u and n_v are equal. If a term $n_r \equiv n_u = n_v$ is defined, Eq. (2a) simplifies to

$$\rho_{\text{spec},i} = \frac{n_r + 1}{8\pi} \frac{(\mathbf{u}_{n,i}^I \cdot \mathbf{u}_h^I)^{n_r}}{\mathbf{u}_{n,i}^I \cdot \mathbf{u}_{\text{sun}}^I + \mathbf{u}_{n,i}^I \cdot \mathbf{u}_{\text{obs}}^I - (\mathbf{u}_{n,i}^I \cdot \mathbf{u}_{\text{sun}}^I)(\mathbf{u}_{n,i}^I \cdot \mathbf{u}_{\text{obs}}^I)} F_{\text{reflect},i} \quad (7)$$

This is referred to as the “Blinn–Phong model.” In this case there is no longer a dependence on $\mathbf{u}_{u,i}$ or $\mathbf{u}_{v,i}$. The variable n_r is a parameter that describes the surface roughness. As $n_r \rightarrow 0$, the surface becomes a uniform reflector, and as $n_r \rightarrow \infty$ it becomes mirror-like.

The diffuse term of the BRDF is given by [13]

$$\rho_{\text{diff},i} = \left(\frac{28R_{\text{diff},i}}{23\pi} \right) (1 - R_{\text{spec},i}) \left[1 - \left(1 - \frac{\mathbf{u}_{n,i}^I \cdot \mathbf{u}_{\text{sun}}^I}{2} \right)^5 \right] \times \left[1 - \left(1 - \frac{\mathbf{u}_{n,i}^I \cdot \mathbf{u}_{\text{obs}}^I}{2} \right)^5 \right] \quad (8)$$

Note that if n and k are used in computation of the specular reflection as in Eq. (3), $R_{\text{spec},i}$ may not be available for that material. As this is required in Eq. (8), it can be approximated by inversion of Schlick's approximation as

$$R_{\text{spec},i} = \frac{F_{\text{reflect},i} - (1 - c)^5}{1 - (1 - c)^5} \quad (9)$$

where $F_{\text{reflect},i}$ and c are computed using Eq. (3). Note that $R_{\text{spec},i}$, as used in the BRDF model, corresponds the specular reflectance at normal incidence. Therefore, $c = 1$ and Eq. (9) simplifies to

$$R_{\text{spec},i} = F_{\text{reflect},i} \quad (10)$$

Furthermore, with $c = 1$ at normal incidence, a and b in Eq. (3) reduce to

$$a = n \quad (11a)$$

$$b = k \quad (11b)$$

Therefore, the specular reflectance reduces to

$$R_{\text{spec},i} = \frac{(n-1)^2 + k^2}{(n+1)^2 + k^2} \quad (12)$$

It is often assumed that diffuse reflection occurs primarily due to surface roughness, and that the surface roughness controls the proportion of light that has specular versus diffuse reflection. However, this is generally a poor characterization. Diffuse reflection occurs primarily due to subsurface scattering of light and subsequent reflection back into the surrounding medium, while specular reflection is reflection of light directly from the surface. This mechanism of diffuse reflection is illustrated in Fig. 2. In a diffuse reflector, the material is polycrystalline, meaning that it is composed of a mosaic of many small, irregularly shaped crystals. When a ray of light hits the surface, part of it will be reflected, while part of it will enter a crystal. At the boundary between crystals, part of the light will reflect off the intercrystal boundary, and part of the light will be transmitted into the adjacent crystal. A significant portion of the light may be absorbed within the crystal as well. Some of the reflected rays will be reflected back out of the surface into the surrounding medium, and the angle that they leave the surface depends on the orientation of the crystal surface that they reflect from as well as the ratio of refractive indices of the material and surrounding medium. Some internally scattered light will strike the surface at an angle greater than the critical angle and will only be reflected back into the material. However, the critical angle corresponds to a reflection parallel to the surface, so reflection out of the surface in any direction is possible given incident rays striking at less than the critical angle. Since the crystalline structure is irregular, the surfaces are randomly oriented, and as such, on the macro scale, reflections will occur in all directions.

Therefore, in order to exhibit diffuse reflection by this method, a material must have a nonzero transmissivity, meaning that some of the incident light must be transmitted through the material. This is the case for most materials, but not for metals. For example, light is transmitted through a plastic surface, but not through a metal surface. Although plastics may appear opaque to the eye, if one takes a thin shaving of plastic and shines light through it, some light would pass through. However, if this were to be repeated with a sheet of aluminum foil, no matter how thin the sheet, light would not pass through. Therefore, metals do not exhibit this mechanism of diffuse reflection. Although it is sometimes assumed that metals also reflect diffusely due to surface roughness, this surface roughness is actually accounted for as n_r , or n_u and n_v , in the specular reflectance term. With low values of n_r , it is entirely possible to have a metal surface that appears to be a diffuse reflector, without actually exhibiting diffuse reflectance. Additionally, transparent materials such as glass or polycarbonate do not have a polycrystalline structure, so they do not reflect diffusely. They do, however, exhibit specular reflections.

The exception to this is that diffuse reflection can also occur due to reflection off multiple facets. If a surface is highly irregular, light may reflect off one microfacet onto another, until it is eventually reflected back toward an observer. This is also considered diffuse reflection, and can occur for any type of material, including metals. If polarization is not of concern, this can be lumped in the $\rho_{\text{diff},i}$ term along with the diffuse reflection due to subsurface scattering without issue. The

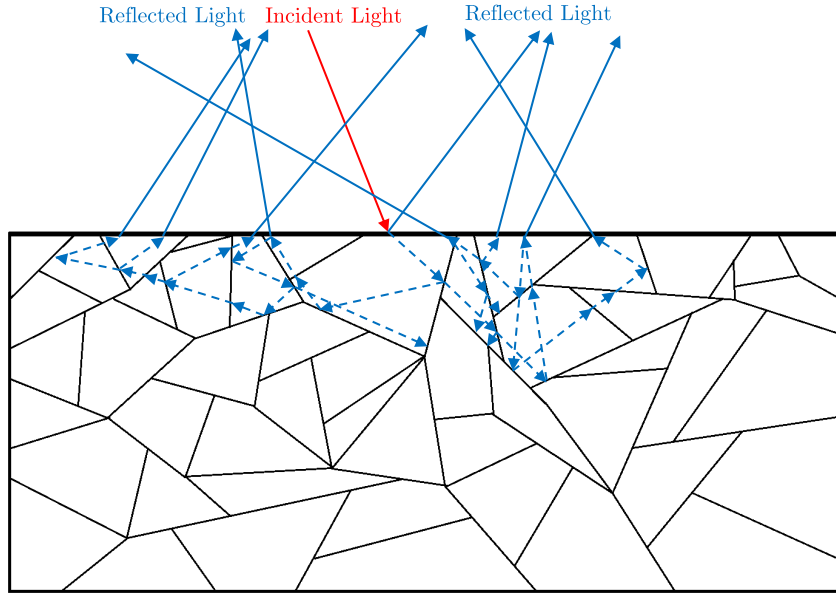


Fig. 2 Diffuse reflection due to subsurface scattering.

impacts on polarized reflections will be discussed as applicable in subsequent sections.

The apparent magnitude of the object is the result of sunlight reflecting off its surfaces along the line-of-sight to an observer. First, the fraction of visible sunlight that strikes an object is computed by

$$F_{\text{sun},i} = C_{\text{sun,vis}} \rho_{\text{total},i} (\mathbf{u}_{n,i}^I \cdot \mathbf{u}_{\text{sun}}^I) \quad (13)$$

where $C_{\text{sun,vis}} = 455 \text{ W/m}^2$ is the power per square meter impinging on a given object due to visible light striking the surface. If either the angle between the surface normal and the observer's direction or the angle between the surface normal and the sun direction is greater than $\pi/2$, then there is no light reflected toward the observer. If this is the case, then the fraction of visible light is set to $F_{\text{sun},i} = 0$.

The fraction of sunlight that strikes a surface that is reflected is given by

$$F_{\text{obs},i} = \frac{F_{\text{sun},i} A_{\text{facet},i} (\mathbf{u}_{n,i}^I \cdot \mathbf{u}_{\text{obs}}^I)}{d^2} \quad (14)$$

where d is the distance from the observer to the object and $A_{\text{facet},i}$ is the area of the facet. The reflected light is now used to compute the apparent brightness magnitude, which is measured by an observer through

$$m_{\text{app}} = -26.7 - 2.5 \log_{10} \left| \sum_{i=1}^N \frac{F_{\text{obs},i}}{C_{\text{sun,vis}}} \right| \quad (15)$$

where -26.7 is the apparent magnitude of the sun, and N is the total number of facets. It is important to note that m_{app} is defined across a certain wavelength range (in this case, the entire visible spectrum). If a multispectral BRDF is considered, then $m_{\text{app},\lambda}$ would be defined for each band λ , and $C_{\text{sun},\lambda}$ would then be the Watts per square meter of sunlight in the wavelength range for that band. An array of measurements for each wavelength range λ would then be available, as opposed to a single scalar.

B. Polarized Light Curve Model

1. Polarized Light Representations

While the scalar m_{app} gives the total intensity of the reflected light, it does not contain information about the light's polarization. However, the Fresnel reflectance of unpolarized light (such as sunlight) will be polarized so long as the vectors $\mathbf{u}_h^I$ and $\mathbf{u}_n^I$ are not aligned. One common example of this phenomenon can be observed by looking at the surface of a body of water or a metal roof while wearing polarized sunglasses. As one rotates their head, the observed intensity of the

specular reflection, or "glare," will change. Since many surfaces of RSOs are relatively flat, they will produce polarized Fresnel reflectance components. Additionally, the diffuse reflection of unpolarized light is polarized, although to a lesser degree than specular reflection. Unlike specular reflection, the polarization of the diffuse reflection depends only on the angle at which the reflection is viewed, and not the angle to the light source. Measures of the polarization can thus provide additional information beyond the scalar magnitude.

Polarized light is most often described using the Stokes vector notation. The Stokes vector is composed of the Stokes parameters, which are defined in terms of the polarization ellipse. Elliptical polarization is the most general form of polarized light. It can be resolved into two linearly polarized waves offset by $\pi/2$ radians (phase quadrature). The projection of these waves on a plane perpendicular to the light's direction of travel will form an ellipse. Figure 3 shows the polarization ellipse.

The polarization ellipse can be expressed mathematically as [16]

$$\frac{e_x^2}{e_{0x}^2} + \frac{e_y^2}{e_{0y}^2} + \frac{-2e_x e_y}{e_{0x} e_{0y}} \cos \phi = \sin^2 \phi \quad (16)$$

The polarization angle, denoted as ψ in Fig. 3, is computed as [16]

$$\tan 2\psi = \frac{2e_{0x} e_{0y} \cos \phi}{e_{0x}^2 - e_{0y}^2} \quad (17)$$

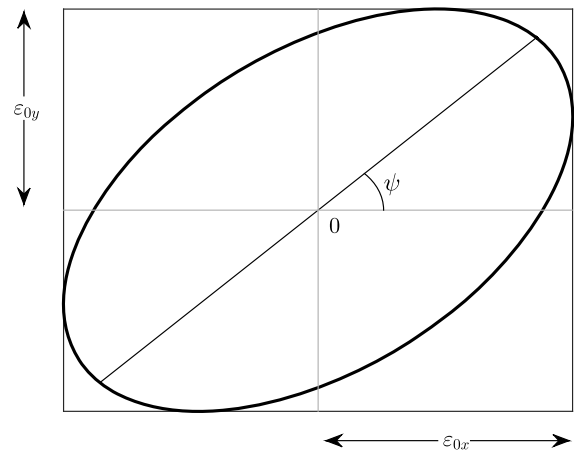


Fig. 3 Polarization ellipse.

Stokes showed that if the time averages of ε_x and ε_y are taken, Eq. (16) becomes

$$(\varepsilon_{0x}^2 + \varepsilon_{0y}^2)^2 = (\varepsilon_{0x}^2 - \varepsilon_{0y}^2)^2 + (2\varepsilon_{0x}\varepsilon_{0y} \cos \phi)^2 + (2\varepsilon_{0x}\varepsilon_{0y} \sin \phi)^2 \quad (18)$$

This formulation breaks down the overall light intensity into its polarization components, and can be expressed in the form

$$S_0^2 = S_1^2 + S_2^2 + S_3^2 \quad (19)$$

where S_0, S_1, S_2 , and S_3 are the Stokes parameters, which are defined as [16]

$$S_0 = \varepsilon_{0x}^2 + \varepsilon_{0y}^2 \quad (20a)$$

$$S_1 = \varepsilon_{0x}^2 - \varepsilon_{0y}^2 \quad (20b)$$

$$S_2 = 2\varepsilon_{0x}\varepsilon_{0y} \cos \phi \quad (20c)$$

$$S_3 = 2\varepsilon_{0x}\varepsilon_{0y} \sin \phi \quad (20d)$$

It is important to note that these parameters are real quantities that have units of intensity or energy in a beam. The variable S_0 describes the total energy in a beam. The variable S_1 describes the amount of linear horizontal or vertical polarization. The variable S_2 describes the amount of $\pm 45^\circ$ linear polarization, and S_3 describes the amount of right- or left-handed circular polarization [16]. Note that Eq. (18) describes fully polarized light, and thus if the light is partially polarized, this will increase the S_0 term such that $S_0^2 > S_1^2 + S_2^2 + S_3^2$. Thus, for describing the generalized case of partially polarized light, the Stokes parameters are not redundant. The Stokes vector S is formed from the Stokes parameters:

$$S = [S_0 \ S_1 \ S_2 \ S_3]^T \quad (21)$$

Note that if $\varepsilon_0^2 \equiv \varepsilon_{0x}^2 + \varepsilon_{0y}^2$, then $S_0 = \varepsilon_0^2$, and the Stokes vector can be written as

$$S = \varepsilon_0^2 \hat{S} \quad (22)$$

where

$$\hat{S} \equiv [1 \ S_1/S_0 \ S_2/S_0 \ S_3/S_0]^T \quad (23)$$

Here, $\hat{S}$ contains the polarization state of the light beam without containing its total intensity. It is also common to refer to this quantity as the Stokes vector [16]. In this work, the terms S and $\hat{S}$ will be used to avoid confusion. The $\hat{S}$ vector can be used to describe a number of important polarization states. Randomly polarized (or “unpolarized”) light has $\hat{S} = [1 \ 0 \ 0 \ 0]^T$. Linear horizontal polarization has $\hat{S} = [1 \ 1 \ 0 \ 0]^T$ and linear vertical polarization has $\hat{S} = [1 \ -1 \ 0 \ 0]^T$. Linear $+45^\circ$ polarization has $\hat{S} = [1 \ 0 \ 1 \ 0]^T$ and linear -45° polarization has $\hat{S} = [1 \ 0 \ -1 \ 0]^T$. Right-handed circular polarization has $\hat{S} = [1 \ 0 \ 0 \ 1]^T$ and left-handed circular polarization has $\hat{S} = [1 \ 0 \ 0 \ -1]^T$.

2. Modeling Polarized Reflections

Given the Stokes vector representation of polarized light, a method by which to compute the polarized reflection of a light beam must be implemented. The Mueller matrix M allows the Stokes vector of an incoming light beam, S_{in} , to be mapped to the Stokes vector of a transmitted or reflected light beam, S_{out} :

$$S_{\text{out}} = MS_{\text{in}} \quad (24)$$

Mueller matrices can be defined for either transmission through a medium or reflection off a surface. The Mueller matrix for polarimetric Fresnel reflection is given by [16]

$$M_R = \frac{1}{2} \begin{bmatrix} R_s + R_p & R_s - R_p & 0 & 0 \\ R_s - R_p & R_s + R_p & 0 & 0 \\ 0 & 0 & 2\text{Re}(r_s r_p^*) & 2\text{Im}(r_s r_p^*) \\ 0 & 0 & -2\text{Im}(r_s r_p^*) & 2\text{Re}(r_s r_p^*) \end{bmatrix} \quad (25)$$

where $r_s = \sqrt{R_s}$; $r_p = \sqrt{R_p}$; Re and Im refer to the real and imaginary parts, respectively; and the superscript “*” denotes the complex conjugate. The variable R_s is the Fresnel reflectance coefficient for radiation perpendicular to the plane of incident flux, and R_p is the Fresnel reflectance coefficient for radiation parallel to the plane of incident flux. Likewise, the Mueller matrix for transmission through a surface is given by [17]

$$M_T = \frac{1}{2} \begin{bmatrix} T_s + T_p & T_s - T_p & 0 & 0 \\ T_s - T_p & T_s + T_p & 0 & 0 \\ 0 & 0 & 2\text{Re}(\tau_s \tau_p^*) & 2\text{Im}(\tau_s \tau_p^*) \\ 0 & 0 & -2\text{Im}(\tau_s \tau_p^*) & 2\text{Re}(\tau_s \tau_p^*) \end{bmatrix} \quad (26)$$

where $\tau_s = \sqrt{T_s}$, $\tau_p = \sqrt{T_p}$, and the transmission is the part of the light that is not reflected, $T = 1 - R$, which can be written componentwise as

$$T_s = 1 - R_s \quad (27a)$$

$$T_p = 1 - R_p \quad (27b)$$

Since sunlight is randomly polarized, it has a Stokes vector of $S_{\text{in}} = [1 \ 0 \ 0 \ 0]^T$. Thus, the Stokes vector of the Fresnel reflection is

$$S_F = [R_s + R_p \ R_s - R_p \ 0 \ 0]^T \quad (28)$$

From this, it is seen that the Fresnel reflectance of randomly polarized light has linear horizontal or vertical polarization, but not $\pm 45^\circ$ or circular polarization. Thus, in the case of reflected sunlight, the measurements are the S_0 and S_1 components of the reflected light. Note that S_0 is the total intensity, as is used in unpolarized light curve analysis. Polarimetric observations add a second measurement, the S_1 component.

Another quantity by which polarization can be represented is the degree of polarization, which gives the fraction of incident light that is polarized. The degree of polarization can be computed from the Stokes parameters as [16]

$$V = \frac{\sqrt{S_1^2 + S_2^2 + S_3^2}}{S_0} \quad (29)$$

For the case of an unpolarized incident beam with S_2 and S_3 components of the reflection being zero, this equation simplifies to

$$V = \frac{S_1}{S_0} = \frac{R_s - R_p}{R_s + R_p} \quad (30)$$

The degree of polarization is commonly used as it allows the amount of polarization to be represented without dependence on the overall intensity S_0 . In some cases, polarized measurements are given as S_0 and V instead of S_0 and S_1 . These measurements can easily be converted into Stokes vectors through Eq. (30).

3. Polarized BRDF Model

From the previous section, it is seen that the Fresnel reflection of unpolarized light results in a polarized reflection. Fresnel reflection occurs in specular reflection. The s and p components of the Fresnel reflection are denoted R_s and R_p and can be computed as [16]

$$R_s = \left| \frac{(m^2 - \sin^2 \theta_i)^{1/2} - \cos \theta_i}{(m^2 - \sin^2 \theta_i)^{1/2} + \cos \theta_i} \right|^2 \quad (31a)$$

$$R_p = \left| \frac{m^2 \cos \theta_i - (m^2 - \sin^2 \theta_i)^{1/2}}{m^2 \cos \theta_i + (m^2 - \sin^2 \theta_i)^{1/2}} \right|^2 \quad (31b)$$

where m denotes the complex index of refraction, which accounts for the index of refraction as well as the extinction coefficient in the form [12]

$$m_i = n_i + ik_i \quad (32)$$

Here, $m \equiv m_t/m_i$ is the relative index of refraction of the reflected medium to the medium of the incident beam and θ_i is the angle of incidence. Note that specular reflection of a mirror facet is reflected only along the angle $\theta_r = \theta_i$ in the plane of incidence. If the surface has roughness and is modeled as a distribution of microfacets, only the microfacets that are oriented such that $\mathbf{u}_n = \mathbf{u}_h$ will have specular reflection visible by the observer. The term that multiplies F_{reflect} in Eq. (2a) defines the distribution of facets oriented in this direction. Thus, it is seen that the degree of polarization is independent from the orientation of the facet. For given locations of the observer, reflecting body, and light source, the angle θ_i is replaced by θ , or the angle between the object-sun and object-observer vectors:

$$\theta = \text{acos}(\mathbf{u}_{\text{sun}}^l \cdot \mathbf{u}_h^l) \quad (33)$$

For conductors (such as metals), k is nonzero, and thus m is complex. Equation (31) can be expanded to [18]

$$R_s = \frac{a^2 + b^2 - 2a \cos \theta + \cos^2 \theta}{a^2 + b^2 + 2a \cos \theta + \cos^2 \theta} \quad (34a)$$

$$R_p = R_s \frac{a^2 + b^2 - 2a \sin \theta \tan \theta + \sin^2 \theta \tan^2 \theta}{a^2 + b^2 + 2a \sin \theta \tan \theta + \sin^2 \theta \tan^2 \theta} \quad (34b)$$

where a and b are as given in Eq. (3). R_s and R_p are the components of the Fresnel reflection in the planes perpendicular and parallel to the plane of the incident flux, respectively. The unpolarized Fresnel reflectance is simply the summation $F_{\text{reflect}} = R_s + R_p$. Note that for modeling polarized reflections, n and k must be known, and the approximation of Eq. (5) cannot be applied.

To extend the BRDF model to account for polarization, the $F_{\text{reflect},i}$ term becomes a vector with s and p components. If only linear vertical/horizontal polarization is considered, this vector has size 2×1 . As this term is carried through the model, the terms $\rho_{\text{spec},i}$, $\rho_{\text{total},i}$, $F_{\text{sun},i}$, $F_{\text{obs},i}$, and m_{app} also become vectors. In the polarized form, Eq. (3) will become

$$\mathbf{F}_{\text{reflect},i} = [R_s \ R_p]^T \quad (35)$$

This fully models the polarization of the specular reflection.

The polarization of the diffuse reflection is now considered. Many references, including [16], assume that the diffuse reflection is mostly or completely depolarizing. For this to be the case, the diffuse reflection would need to be primarily due to reflection off multiple surface facets, not subsurface scattering. This is typically the case for many diffuse reflections observed in remote sensing, where surfaces are not smooth on a macro scale, such as aerial imagery of foliage or grass.

However, if the surface is artificial, it will likely be relatively smooth and most of the diffuse reflection will be due to subsurface scattering. Light will be transmitted out of the surface as shown in

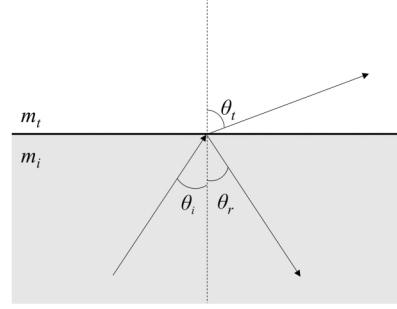


Fig. 4 Transmission from diffuse surface.

Fig. 4. The Fresnel equations are considered for this case. In the general form, these are given as

$$R_s = \left[\frac{m_i \cos \theta_i - m_t \cos \theta_t}{m_i \cos \theta_i + m_t \cos \theta_t} \right]^2 \quad (36a)$$

$$R_p = \left[\frac{m_i \cos \theta_i - m_t \cos \theta_t}{m_i \cos \theta_i + m_t \cos \theta_t} \right]^2 \quad (36b)$$

Snell's law gives the relation between the incident and reflected beams:

$$m_i \sin \theta_i = m_t \sin \theta_t \quad (37)$$

Equation (37) is typically solved for θ_t in terms of θ_i and substituted into the Fresnel equations, which results in Eq. (31). However, in this case, it is solved for θ_i in terms of θ_t . Additionally, the order of m_i and m_t are reversed compared to the case where light is reflecting from outside the material, so the refractive index is replaced with $1/m$. This results in

$$R_s = \left[\frac{(m^2 - \sin^2 \theta_t)^{1/2} - \cos \theta_t}{(m^2 - \sin^2 \theta_t)^{1/2} + \cos \theta_t} \right]^2 \quad (38a)$$

$$R_p = \left[\frac{(m^2 - \sin^2 \theta_t)^{1/2} - m^2 \cos \theta_t}{(m^2 - \sin^2 \theta_t)^{1/2} + m^2 \cos \theta_t} \right]^2 \quad (38b)$$

Note that this now is of the same form as Eq. (36). In terms of the unit vectors that form the light curve model, θ_t is expressed as

$$\theta_t = \text{acos}(\mathbf{u}_{n,i}^l \cdot \mathbf{u}_{\text{obs}}^l) \quad (39)$$

Therefore, Eq. (38) can be written as

$$R_s = \left[\frac{\sqrt{m^2 - (1 - \mathbf{u}_{n,i}^l \cdot \mathbf{u}_{\text{obs}}^l)^2} - \mathbf{u}_{n,i}^l \cdot \mathbf{u}_{\text{obs}}^l}{\sqrt{m^2 - (1 - \mathbf{u}_{n,i}^l \cdot \mathbf{u}_{\text{obs}}^l)^2} + \mathbf{u}_{n,i}^l \cdot \mathbf{u}_{\text{obs}}^l} \right]^2 \quad (40a)$$

$$R_p = \left[\frac{\sqrt{m^2 - (1 - \mathbf{u}_{n,i}^l \cdot \mathbf{u}_{\text{obs}}^l)^2} - m^2 (\mathbf{u}_{n,i}^l \cdot \mathbf{u}_{\text{obs}}^l)}{\sqrt{m^2 - (1 - \mathbf{u}_{n,i}^l \cdot \mathbf{u}_{\text{obs}}^l)^2} + m^2 (\mathbf{u}_{n,i}^l \cdot \mathbf{u}_{\text{obs}}^l)} \right]^2 \quad (40b)$$

Here, R_s and R_p are the components of the light that are internally reflected back into the material. To determine the diffuse reflectance, the components of the light that are transmitted back into the viewing medium are desired. These transmissivity components are defined in Eq. (27). The light that is internally incident on the surface is not equal intensity to the light that externally strikes the surface, as the material will absorb some of the light, and part of the light will pass through the material. The amount of light that is transmitted out of the surface depends on a number of factors, including the material's

absorptivity and crystal structure. However, since the diffusely reflected light does not depend on the original angle of incidence from the light source, it can be captured as the scalar R_{diff} , as in Eq. (8). In the calculation of the specular reflection, the index of refraction and full Fresnel equations are used, instead of the R_{spec} value and Schlick's approximation. For consistency with this convention, it may be desirable to express R_{diff} in terms of the more fundamental material properties. One approximation of R_{diff} is the Schuster–Kubelka–Munk model [19], which gives

$$R_{\text{diff}} = \frac{s}{\alpha + s + \sqrt{\alpha(\alpha + 2s)}} \quad (41)$$

where α is the absorption coefficient and s is the scattering coefficient. The absorption coefficient can be computed from the extinction coefficient as

$$\alpha = \frac{4\pi f k}{c} \quad (42)$$

where f is the frequency of the light being reflected. However, the scattering coefficient s is independent of the index of refraction and extinction coefficient, and therefore adds another parameter to the model. The scattering coefficient is wavelength dependent, and unfortunately for many materials it is not well-characterized. Often, laboratory measurements of R_{diff} are more readily available.

The diffuse reflection can be separated into polarized components as

$$\rho_{\text{diff},i,s} = \frac{T_s}{T_s + T_p} k_{\text{sub}} \rho_{\text{diff},i} \quad (43a)$$

$$\rho_{\text{diff},i,p} = \frac{T_p}{T_s + T_p} k_{\text{sub}} \rho_{\text{diff},i} \quad (43b)$$

where $\rho_{\text{diff},i}$ is given in Eq. (8). The variable k_{sub} denotes the fraction of the diffuse reflection that is due to subsurface scattering, and not reflection off multiple facets. In practice, this quantity will depend on the viewing geometry, including the angle of incidence and the angle to the observer, as well as surface roughness, crystalline structure, and other factors. As such, this cannot be modeled analytically. Due to the number of variables that affect this and the lack of usefulness to most conventional BRDF applications such as computer graphics, empirical models do not exist. As such, microfacet models must be used for accurate simulation of diffuse polarization. In microfacet models, surfaces are generated as a collection of microfacets, and the reflection of light off each microfacet, as well as the interaction between multiple microfacets, is modeled. Polarized light curves of spacecraft have been simulated using microfacet models in [12]. However, microfacet models are of limited use in efforts to characterize RSO parameters. While it is computationally expensive to run microfacet models in filtering applications, this process can be easily parallelized, which lends itself well to cluster computing. However, since the microfacet structure of the object itself is not known, there is no model to use in a filter. The same material may have vastly different microfacet structures depending on how it was manufactured and processed, so using a model measured in a laboratory may not bear significant semblance to the actual RSO. While it is possible that many microfacet models could be run in a multiple-model estimator, the relatively small amount of information available from the light curve would make this an ill-conditioned problem.

However, for the case of artificial surfaces, it can be accurate enough to assume that the surface is smooth enough that all diffuse reflection is due to subsurface scattering, in which case $k_{\text{sub}} = 1$. This assumption is used in [20], which estimates the surface normal direction from diffuse polarization. In this work real polarization images are taken of several porcelain and rubber objects, and surface orientation recovery is successfully accomplished using these real images. This shows a real case where the multifacet reflections can be neglected. Nevertheless, this work also acknowledges that multifacet

scattering has a depolarization effect, which results in underestimation of the zenith angle. For the subsequent analysis in this work, it is assumed that all diffuse reflection is due to subsurface scattering.

4. Multilayer Transparent Materials

A more complicated, yet relevant, reflection case involves a transparent material such as glass covering another material, as shown in Fig. 5. This is common on spacecraft, where solar panels typically consist of gallium arsenide solar cells with a glass cover to protect the cells. In this case, the glass is amorphous and does not diffusely reflect light. However, specular reflections occur both from reflection off the top of the glass as well as reflection off the covered material. Because the specular reflection refracts back into the originating medium, both specular reflections will be in the same direction. However, if the cover material has nonnegligible thickness, the point of origination will be different. If viewed from far away, as in the case of remote observations of spacecraft, this image offset becomes negligible.

Diffuse reflections occur because of the subsurface scattering of the covered material; however, the degree of polarization of the diffuse reflection depends on the cover glass as well. Although one might expect transmittance out of the covering material to have the opposite polarizing effect as transmission into the material, this is not the case. When the light passes through an interface from a lower to a higher index of refraction, more s -polarized light is reflected than p -polarized light. Therefore, the light transmitted into the material will have a greater degree of p polarization. When the light exits the higher-index medium, the internal reflection also has a greater degree of s polarization than p polarization, so transmitted light becomes further p polarized. Therefore, the presence of the glass increases the p polarization.

The Mueller matrix formulation allows the polarizing effect of the multilayer surface to be easily computed. Since the illumination of the object is assumed to be sunlight, only the first two components of the Stokes vector are of interest, as shown above. For succinctness, truncated 2×1 Stokes vectors and 2×2 Mueller matrices are used here without loss of generality. Let S_0 be the Stokes vector of the unpolarized incident light, denoted by

$$S_0 = [1 \quad 0]^T \quad (44)$$

Then, let $M_{T,0 \rightarrow 1}$ be the Mueller matrix of transmission from medium 0 to medium 1:

$$M_{T,0 \rightarrow 1} = \frac{1}{2} \begin{bmatrix} T_{s,0 \rightarrow 1} + T_{p,0 \rightarrow 1} & T_{s,0 \rightarrow 1} - T_{p,0 \rightarrow 1} \\ T_{s,0 \rightarrow 1} - T_{p,0 \rightarrow 1} & T_{s,0 \rightarrow 1} + T_{p,0 \rightarrow 1} \end{bmatrix} \quad (45)$$

Also, $M_{R,2}$ denotes the Mueller matrix for reflection off surface 2:

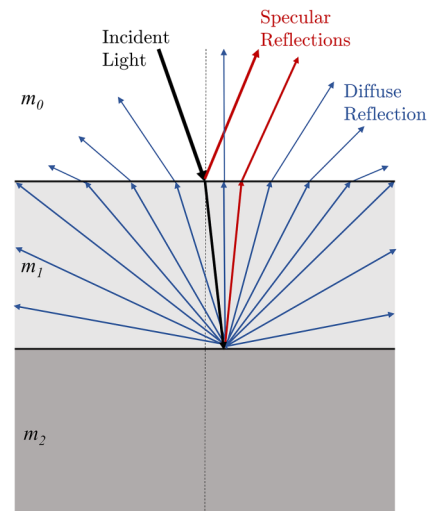


Fig. 5 Multilayer reflections with cover glass.

$$M_{R,2} = \frac{1}{2} \begin{bmatrix} R_{s,2} + R_{p,2} & R_{s,2} - R_{p,2} \\ R_{s,2} - R_{p,2} & R_{s,2} + R_{p,2} \end{bmatrix} \quad (46)$$

Finally, $M_{T,1 \rightarrow 0}$ denotes the Mueller matrix for transmission from medium 1 back into medium 0:

$$M_{T,1 \rightarrow 0} = \frac{1}{2} \begin{bmatrix} T_{s,1 \rightarrow 0} + T_{p,1 \rightarrow 0} & T_{s,1 \rightarrow 0} - T_{p,1 \rightarrow 0} \\ T_{s,1 \rightarrow 0} - T_{p,1 \rightarrow 0} & T_{s,1 \rightarrow 0} + T_{p,1 \rightarrow 0} \end{bmatrix} \quad (47)$$

Then, the Stokes vector of the outgoing light back into medium 0 is

$$\mathbf{S}_{\text{out}} = M_{T,1 \rightarrow 0} M_{R,2} M_{T,0 \rightarrow 1} \mathbf{S}_0 \quad (48)$$

III. System Dynamics and Estimation

Estimation of attitude and size scale-factors is accomplished using an unscented Kalman filter (UKF) [21], and surface materials and shapes are estimated using adaptive estimation methods. The global parameterization of the attitude in the UKF is the quaternion [22], whereas a minimal parameterization involving the generalized Rodrigues parameters (GRPs) is used to define the local error [23]. Quaternions are the global parameterization of choice because their kinematics are free of singularities. The representation of the attitude error as a GRP is useful for the propagation and update steps of the attitude covariance because the structure of the UKF can be used directly. A complete explanation of the quaternion and its mapping to GRPs is provided in [23].

The rotational dynamics are given by the coupled first-order differential equations:

$$\dot{\mathbf{q}}_I^B = \frac{1}{2} \Xi(\mathbf{q}_I^B) \boldsymbol{\omega}_{B/I}^B = \frac{1}{2} \Omega(\boldsymbol{\omega}_{B/I}^B) \mathbf{q}_I^B \quad (49a)$$

$$\dot{\boldsymbol{\omega}}_{B/I}^B = J_{\text{RSO}}^{-1} (\mathbf{T}_{\text{ext}}^B - [\boldsymbol{\omega}_{B/I}^B \times] J_{\text{RSO}} \boldsymbol{\omega}_{B/I}^B) \quad (49b)$$

where $\mathbf{q}_I^B$ is the quaternion representing the rotation from inertial to body coordinates; $\boldsymbol{\omega}_{B/I}^B$ is the angular velocity of the object with respect to the inertial frame, expressed in body coordinates; J_{RSO} is the inertia matrix of the object; and $\mathbf{T}_{\text{ext}}^B$ is the net external torque acting on the object, expressed in body coordinates.

Multiple-model adaptive estimation (MMAE) is a recursive algorithm that uses a bank of estimators, each dependent on a particular hypothesis, to determine an estimate based upon an unknown physical process under consideration. In particular, the hypotheses can correspond to different mathematical models of the same physical process or of the same model but dependent upon different constants or model parameters. Each filter in bank uses an UKF that employs different material properties for state estimation. A random-walk process noise component is used to account for uncertainty in the inertia matrix and external torque in the UKF. The MMAE process provides the real-time probability that a particular filter is the one that best matches the data, thus allowing a mechanism to estimate the material properties and shape of an RSO from polarized measurements. A detailed derivation of the estimation process can be found in [24].

A. Sensor Modeling

The measurements are taken to be the Stokes vector, which is limited to the nonzero (S_0 and S_1) parameters; hence

$$\tilde{\mathbf{y}}_k = [\hat{S}_0 \ \hat{S}_1]^T + \mathbf{v}_k \quad (50)$$

where $\tilde{\mathbf{y}}_k$ is the measurement used in the UKF, and $\mathbf{v}_k$ is the measurement noise, which is assumed to be a zero-mean Gaussian white-noise process with covariance R_k . It is assumed that the polarized light is measured by a micropolarizer array in which each pixel measures a different polarization direction. The raw measurement from a Charge-Coupled Device (CCD) is the electron count n_e and is expressed as

$$\tilde{n}_e = n_e + \eta_d \Delta t + \eta_r \quad (51)$$

where n_e is the true electron count and $\tilde{n}_e$ is the measured electron count, η_d is the dark current noise, η_r is the read noise, and Δt is the exposure time of the image. The spectral density of the noise in the electron count, σ_{n_e} , can be determined as a function of the independent dark current noise and read noise by evaluating for the i th pixel:

$$\sigma_{n_e}^2 = \mathbb{E}\{(\tilde{n}_{e,i} - \mathbb{E}\{\tilde{n}_{e,i}\})(\tilde{n}_{e,i} - \mathbb{E}\{\tilde{n}_{e,i}\})\} = \sigma_d^2 \Delta t^2 + \sigma_r^2 \quad (52)$$

where σ_d and σ_r are the spectral densities of η_d and η_r , respectively, and $\mathbb{E}$ denotes the expected value.

The electron count in a charge well is related to the number of photons striking the charge well by the quantum efficiency Q , which is typically given in units of electrons per photon. The number of photons is given by

$$p = \frac{n_e}{Q \Delta t} \quad (53)$$

The energy per photon is given by

$$E = \frac{hc}{\lambda} \quad (54)$$

where $h = 6.626 \times 10^{-34}$ m² kg/s is Planck's constant, $c = 2.998 \times 10^8$ m/s is the speed of light, and λ is the wavelength of the photon. The measured intensity of the light is expressed as $I = pE$, where I has units of power. If A is the area of the lens or mirror gathering the light that is focused onto the CCD, then the reflected irradiance is

$$F_{\text{obs}} = \frac{I}{A} = \frac{4I}{\pi d^2} = \frac{4hc n_e}{Q \Delta t \pi d^2} \quad (55)$$

The Stokes parameter measurements are now given by

$$S_0 = F_{\text{obs},s} + F_{\text{obs},p} = \frac{4hc}{\lambda \pi d^2 Q \Delta t} (n_{e,s} + n_{e,p}) \quad (56a)$$

$$S_1 = F_{\text{obs},s} - F_{\text{obs},p} = \frac{4hc}{\lambda \pi d^2 Q \Delta t} (n_{e,s} - n_{e,p}) \quad (56b)$$

Here, $n_{e,s}$ and $n_{e,p}$ are the number of electrons measured by the horizontally and vertically polarized pixels, respectively. The uncertainty in the $n_{e,s} - n_{e,p}$ term can be found by taking

$$\begin{aligned} \sigma_{n_{e,s}-n_{e,p}}^2 &= \mathbb{E}\{(\tilde{n}_{e,s} - \tilde{n}_{e,p} - \mathbb{E}\{\tilde{n}_{e,s} - \tilde{n}_{e,p}\})(\tilde{n}_{e,s} - \tilde{n}_{e,p} \\ &\quad - \mathbb{E}\{\tilde{n}_{e,s} - \tilde{n}_{e,p}\})\} \\ &= \mathbb{E}\{(n_{d,s} \Delta t + n_{r,s} - n_{d,p} \Delta t - n_{r,p})^2\} \end{aligned} \quad (57)$$

where $n_{d,s}$ and $n_{d,p}$ are the dark noise in the s and p pixels, respectively, and $n_{r,s}$ and $n_{r,p}$ are the read noise. Since the noise between the two pixels is uncorrelated, this becomes

$$\begin{aligned} \sigma_{n_{e,s}-n_{e,p}}^2 &= \mathbb{E}\{n_{d,s}^2 \Delta t^2 + n_{r,s}^2 + n_{d,p}^2 \Delta t^2 + n_{r,p}^2\} \\ &= \mathbb{E}\{n_{d,s}^2 \Delta t^2 + n_{r,s}^2\} + \mathbb{E}\{n_{d,p}^2 \Delta t^2 + n_{r,p}^2\} \end{aligned} \quad (58)$$

Likewise, the result for $n_{e,s} + n_{e,p}$ is the same. Noting that the readings from the s and p pixels have uncertainties given by Eq. (52), it is seen that

$$\sigma_{n_{e,s}+n_{e,p}} = \sigma_{n_{e,s}-n_{e,p}} = \sqrt{2} \sigma_{n_e} \quad (59)$$

Taking the partials of S_0 and S_1 with respect to these gives

$$\frac{\partial S_0}{\partial (n_{e,s} + n_{e,p})} = \frac{\partial S_1}{\partial (n_{e,s} - n_{e,p})} = \frac{4hc}{\lambda \pi d^2 Q \Delta t} \quad (60)$$

Therefore, the uncertainty in the Stokes parameters is

$$\sigma_{S_0} = \sigma_{S_1} = \frac{4\sqrt{2}hc}{\lambda\pi d^2 Q \Delta t} \sigma_{n_e} \quad (61)$$

Note that, while both Stokes parameters will have the same uncertainty, the magnitude of S_1 is always lower than S_0 for cases of partial polarization, so the signal-to-noise ratio of S_1 will always be lower than S_0 . The off-diagonal terms in the covariance matrix R_k are evaluated as $\mathbb{E}\{(\tilde{n}_{e,s} + \tilde{n}_{e,p} - \mathbb{E}\{\tilde{n}_{e,s} + \tilde{n}_{e,p}\})(\tilde{n}_{e,s} - \tilde{n}_{e,p} - \mathbb{E}\{\tilde{n}_{e,s} - \tilde{n}_{e,p}\})\}$, which after simplification are determined to be zero. Therefore, the covariance matrix R_k is

$$R_k = \begin{bmatrix} \sigma_{S_0}^2 & 0 \\ 0 & \sigma_{S_1}^2 \end{bmatrix} = \frac{32h^2c^2}{\lambda^2\pi^2d^4Q^2(\Delta t)^2} \begin{bmatrix} \sigma_{n_e}^2 & 0 \\ 0 & \sigma_{n_e}^2 \end{bmatrix} \quad (62)$$

B. Magnitude Calibration Accuracy

The prior section has addressed the expected measurement accuracy due to sensor noise. However, this assumes that the intensity measurements are perfectly calibrated, which will not be the case in reality. Photometric calibration is typically performed by measuring the brightness of a reference star or stars, which have a known magnitude. For an Earth-based sensor, the apparent magnitude is affected by the atmospheric extinction and air mass [25]. The atmospheric extinction coefficient and air mass depend on specific atmospheric conditions and experience variation over time due to temperature, humidity, cloud cover, other weather, and external factors. While the calibration will be accurate over short periods, it will not remain so over the course of longer periods. For observation periods that are hours long, the calibration will likely change over this period. As such, calibration is typically performed over the entire measurement period using background stars available in the images. If suitable background stars are unavailable when pointing at the target, the sensor is periodically pointed toward a reference star to update the calibration.

Magnitude calibration accuracy of ± 0.04 magnitude at magnitude 14.25 has been shown in Refs. [26,27]. This equates to a $\sigma_{F_{\text{obs}}} = 7.0 \times 10^{-16}$ W. Therefore, there may be a limit to the useful aperture, beyond which there are diminishing returns. It is noted that this calibration error will likely not be zero-mean Gaussian but will likely have some bias component that will drift over time.

For a polarized sensor, the polarized measurement will be unaffected by the magnitude calibration. Since the calibration errors will be applicable to the entire sensor, they will not affect the difference in brightness between two pixels. Therefore, the S_1 Stokes parameter will not be affected by the magnitude calibration accuracy, while the S_0 parameter that measures the overall intensity will. However, Rayleigh scattering in the atmosphere causes the sky to be polarized. This polarization is due to the scattering of light away from its direction of incidence; since the RSOs are observed directly along the line-of-sight vector, there is no meaningful scattering of their reflection being observed and therefore no change to the polarization state. However, the scattering of light from other sources, such as moonlight, contributes error to both the measured intensity and polarization. While the scattered light may have a high degree of polarization, the total amount of polarization is low if the intensity of the scattered light is low compared to the intensity of the light reflected from the RSO. If the worst-case scenario of a full moon at zenith is assumed, with the RSO at the point in the sky with the greatest polarization, a magnitude 14 object observed with 0.25 arc-sec pixels would have an error in the degree of polarization of approximately 0.001. This scattering contribution could be mostly calibrated out by measuring the polarization in adjacent, star-free pixels. For the purposes of this work, an error of 0.0001 in the degree of polarization is assumed. This is a small impact compared to the magnitude uncertainty through the atmosphere.

IV. Observability Analysis

In this section an observability analysis of the attitude and surface material from polarized light curves is shown. Also, the resulting information ratio is shown when the S_1 Stokes parameter is included compared to just the S_0 parameter.

A. Surface Parameter Observability from Specular Reflection

Unpolarized, monochromatic light curves do not allow for surface-material determination, as the size of an object cannot be decoupled from its surface reflectivity. It is seen in the polarized BRDF model that there is a dependence of the polarization on the n and k surface parameters. Therefore, it is desired to investigate whether n and k are directly observable from the polarized measurements, which would allow surface materials to be easily identified. The partials of a and b in Eq. (3) with respect to n and k are

$$\frac{\partial a}{\partial n} = \frac{n[p^{-1}(n^2 + k^2 - \sin^2\theta) + 1]}{4\sqrt{p + n^2 - k^2 - \sin^2\theta}} \quad (63a)$$

$$\frac{\partial a}{\partial k} = \frac{k[p^{-1}(n^2 + k^2 - \sin^2\theta) - 1]}{4\sqrt{p + n^2 - k^2 - \sin^2\theta}} \quad (63b)$$

$$\frac{\partial b}{\partial n} = \frac{n[p^{-1}(n^2 + k^2 - \sin^2\theta) - 1]}{4\sqrt{p + n^2 - k^2 - \sin^2\theta}} \quad (63c)$$

$$\frac{\partial b}{\partial k} = \frac{k[p^{-1}(n^2 + k^2 - \sin^2\theta) + 1]}{4\sqrt{p + n^2 - k^2 - \sin^2\theta}} \quad (63d)$$

The partials of the Fresnel coefficients R_s and R_p with respect to a and b from Eq. (4) are

$$\frac{\partial R_s}{\partial a} = \frac{4 \cos \theta (a^2 - b^2 - \cos^2 \theta)}{(a^2 + b^2 + 2a \cos \theta + \cos^2 \theta)^2} \quad (64a)$$

$$\frac{\partial R_s}{\partial b} = \frac{8ab \cos \theta}{(a^2 + b^2 + 2a \cos \theta + \cos^2 \theta)^2} \quad (64b)$$

$$\begin{aligned} \frac{\partial R_p}{\partial a} &= R_s \frac{4 \sin \theta \tan \theta (a^2 - b^2 - \sin^2 \theta \tan^2 \theta)}{(a^2 + b^2 + 2a \sin \theta \tan \theta + \sin^2 \theta \tan^2 \theta)^2} \\ &\quad - \frac{\partial R_s}{\partial a} \frac{4a \sin \theta \tan \theta}{a^2 + b^2 + 2a \sin \theta \tan \theta + \sin^2 \theta \tan^2 \theta} \end{aligned} \quad (64c)$$

$$\begin{aligned} \frac{\partial R_p}{\partial b} &= R_s \frac{8ab \sin \theta \tan \theta}{(a^2 + b^2 + 2a \sin \theta \tan \theta + \sin^2 \theta \tan^2 \theta)^2} \\ &\quad - \frac{\partial R_s}{\partial b} \frac{4a \sin \theta \tan \theta}{a^2 + b^2 + 2a \sin \theta \tan \theta + \sin^2 \theta \tan^2 \theta} \end{aligned} \quad (64d)$$

If n and k are observable from the Stokes vector, they must be observable from the Fresnel reflectance F_{reflect} . In this case, choosing F_{reflect} as the measurement from which to assess the observability allows the contributions of the facet orientations to be ignored. Then, the partials of the measurement with respect to n and k are given by the matrix

$$H = \begin{bmatrix} \frac{\partial R_s}{\partial a} \frac{\partial a}{\partial n} + \frac{\partial R_s}{\partial b} \frac{\partial b}{\partial n} & \frac{\partial R_s}{\partial a} \frac{\partial a}{\partial k} + \frac{\partial R_s}{\partial b} \frac{\partial b}{\partial k} \\ \frac{\partial R_p}{\partial a} \frac{\partial a}{\partial n} + \frac{\partial R_p}{\partial b} \frac{\partial b}{\partial n} & \frac{\partial R_p}{\partial a} \frac{\partial a}{\partial k} + \frac{\partial R_p}{\partial b} \frac{\partial b}{\partial k} \end{bmatrix} \quad (65)$$

If Eq. (64) is substituted into Eq. (65) and simplified, then it is found that $\text{rank}(H^T H) = 1$, so there does not exist direct observability of n and k . Therefore, n and k cannot be directly estimated in a filter. However, the inclusion of the S_1 Stokes parameter still provides information about n and k . Since there is a finite set of materials (i.e., the nk space is not spanned continuously), MMAE may be used to estimate the

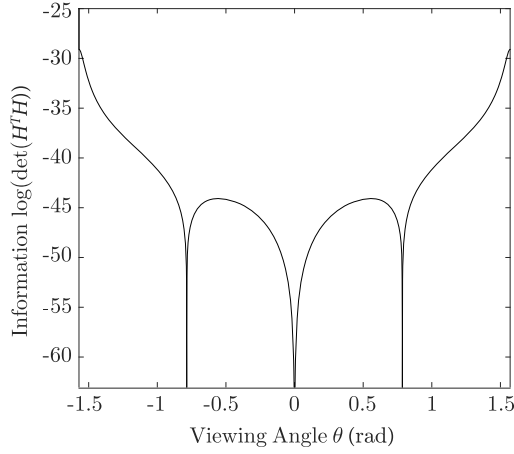


Fig. 6 Magnitude of information of n and k contributed by Stokes vector for aluminum ($n = 1.3$, $k = 7.6$).

material composition of the RSO. The information is quantified as $\log[\det(H^T H)]$ as in [5] and is plotted against the angle between the object–sun and object–observer vectors, θ , in Fig. 6. It is noted that there are asymptotes approaching negative infinity at $\theta = \pm\pi/2$ and $\theta = 0$. This indicates that no information about n or k is contributed at these angles. This occurs because in the Fresnel equations, light reflected at $\theta = 0$ has no change to its polarization, and light reflected at $\theta = \pm\pi/2$ undergoes a $\pi/2$ rotation of the polarization. Since the incident light is unpolarized sunlight, both of these cases have $S_1 = 0$ and no information about n or k is contributed. It is seen that as the grazing incidence $\theta = \pm\pi$ is approached, the information ratio increases.

B. Attitude Observability from Diffuse Reflection

It is seen from the BRDF model that attitude does not affect the polarized signature from pure specular reflection. However, it does affect the polarization of the diffuse reflection. To assess the potential value of the polarization on attitude determination, the attitude observability from polarized measurements is determined. Two approaches to observability analysis will be considered here. First, the observability of desired parameters will be assessed in order to determine whether observability exists, and therefore whether they can be estimated directly in a filter. Second, the amount of information that a particular measurement set contributes to a system will be assessed, whether each parameter is directly observable or not. This is analogous to quantifying how observable a particular system state is. Discussion of the particular measures of observability used will follow in this section. In each case, observability depends on how much a change to a system state affects the observed output, so the first necessary step is to calculate the partials of the observations with respect to the system states. The partials of the diffuse reflectance components $\rho_{\text{diff},s}$ and $\rho_{\text{diff},p}$ with respect to the attitude $\delta\alpha$ are required. Since the partials would be very long if written as single expressions, they are expressed here in terms of intermediate quantities. If the reflectances of Eq. (40) are written of the form

$$R_s = \left[\frac{r_{s,\text{num}}}{r_{s,\text{den}}} \right]^2 \quad (66a)$$

$$R_p = \left[\frac{r_{p,\text{num}}}{r_{p,\text{den}}} \right]^2 \quad (66b)$$

and as per Eq. (27), $T_s = 1 - R_s$ and $T_p = 1 - R_p$, then through application of the chain rule, the partials of T_s and T_p with respect to $\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T$ are

$$\frac{\partial T_s}{\partial(\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T)} = -2 \frac{r_{s,\text{num}}}{r_{s,\text{den}}^3} \left[r_{s,\text{den}} \frac{\partial r_{s,\text{num}}}{\partial(\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T)} - r_{s,\text{num}} \frac{\partial r_{s,\text{den}}}{\partial(\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T)} \right] \quad (67a)$$

$$\frac{\partial T_p}{\partial(\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T)} = -2 \frac{r_{p,\text{num}}}{r_{p,\text{den}}^3} \left[r_{p,\text{den}} \frac{\partial r_{p,\text{num}}}{\partial(\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T)} - r_{p,\text{num}} \frac{\partial r_{p,\text{den}}}{\partial(\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T)} \right] \quad (67b)$$

where

$$\frac{\partial r_{s,\text{num}}}{\partial(\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T)} = \frac{1 - (\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T)}{\sqrt{m^2 - (1 - \mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T)^2}} - 1 \quad (68a)$$

$$\frac{\partial r_{s,\text{den}}}{\partial(\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T)} = \frac{1 - (\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T)}{\sqrt{m^2 - (1 - \mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T)^2}} + 1 \quad (68b)$$

$$\frac{\partial r_{p,\text{num}}}{\partial(\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T)} = \frac{1 - \mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T}{\sqrt{m^2 - (1 - \mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T)^2}} - m^2 \quad (68c)$$

$$\frac{\partial r_{p,\text{den}}}{\partial(\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T)} = \frac{1 - \mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T}{\sqrt{m^2 - (1 - \mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T)^2}} + m^2 \quad (68d)$$

Since the only attitude dependence of these terms is through $\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T$, the partials with respect to the attitude error state $\delta\alpha$ are

$$\frac{\partial T_s}{\partial\delta\alpha} = \frac{\partial T_s}{\partial(\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T)} \frac{\partial(\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T)}{\partial\delta\alpha} \quad (69a)$$

$$\frac{\partial T_p}{\partial\delta\alpha} = \frac{\partial T_p}{\partial(\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T)} \frac{\partial(\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T)}{\partial\delta\alpha} \quad (69b)$$

where

$$\frac{\partial(\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T)}{\partial\delta\alpha} = \mathbf{u}_{\text{obs}}^T \cdot \frac{\partial\mathbf{u}_{n,i}^T}{\partial\delta\alpha}$$

The partials of the diffuse reflectance components in Eq. (43) are then taken as

$$\frac{\partial\rho_{\text{diff},s}}{\partial\delta\alpha} = \frac{1}{(T_s + T_p)^2} \left(T_p \frac{\partial T_s}{\partial\delta\alpha} - T_s \frac{\partial T_p}{\partial\delta\alpha} \right) \rho_{\text{diff}} + \frac{T_s}{T_s + T_p} \frac{\partial\rho_{\text{diff}}}{\partial\delta\alpha} \quad (70a)$$

$$\frac{\partial\rho_{\text{diff},p}}{\partial\delta\alpha} = \frac{1}{(T_s + T_p)^2} \left(T_s \frac{\partial T_p}{\partial\delta\alpha} - T_p \frac{\partial T_s}{\partial\delta\alpha} \right) \rho_{\text{diff}} + \frac{T_p}{T_s + T_p} \frac{\partial\rho_{\text{diff}}}{\partial\delta\alpha} \quad (70b)$$

The partial of the diffuse reflectance is

$$\frac{\partial\rho_{\text{diff},i}}{\partial\delta\alpha} = \frac{5k_{2,i}}{2} \left\{ \left(1 - \frac{\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{sun}}^T}{2} \right)^4 \frac{\partial(\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{sun}}^T)}{\partial\delta\alpha} \left[1 - \left(1 - \frac{\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T}{2} \right)^5 \right] + \left(1 - \frac{\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T}{2} \right)^4 \frac{\partial(\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T)}{\partial\delta\alpha} \left[1 - \left(\frac{\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{sun}}^T}{2} \right)^5 \right] \right\} \quad (71)$$

where

$$k_{2,i} = \left(\frac{28R_{\text{diff},i}}{23\pi} \right) (1 - R_{\text{spec},i}) \quad (72)$$

The partial of the total intensity in the light curve model is given by

$$\frac{\partial F_{\text{obs},i}}{\partial\delta\alpha} = \frac{F_{\text{obs},i}}{F_{\text{sun},i}} \frac{\partial\rho_{\text{total},i}}{\partial\delta\alpha} + \frac{F_{\text{obs},i}}{\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T} \frac{\partial(\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{sun}}^T)}{\partial\delta\alpha} \quad (73)$$

Since all reflection is assumed diffuse, then $\rho_{\text{total},i} = \rho_{\text{diff},i}$. The Stokes parameters for this case are $S_0 = \sum_{i=1}^N (F_{\text{obs},s,i} + F_{\text{obs},p,i})$ and $S_1 = \sum_{i=1}^N (F_{\text{obs},s,i} - F_{\text{obs},p,i})$. Therefore, the partials of the Stokes parameters with respect to the attitude error state are

$$\frac{\partial S_0}{\partial \delta \alpha} = \sum_{i=1}^N \frac{\partial F_{\text{obs},i}}{\partial \delta \alpha} \quad (74a)$$

$$\begin{aligned} \frac{\partial S_1}{\partial \delta \alpha} = \sum_{i=1}^N & \left[\frac{A_{\text{facet},i} (\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T)}{d^2} \left(\frac{\partial \rho_{\text{diff},s,i}}{\partial \delta \alpha} - \frac{\partial \rho_{\text{diff},p,i}}{\partial \delta \alpha} \right) \right. \\ & \left. + \frac{S_1}{\partial \mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{obs}}^T} \frac{\partial (\mathbf{u}_{n,i}^T \cdot \mathbf{u}_{\text{sun}}^T)}{\partial \delta \alpha} \right] \end{aligned} \quad (74b)$$

From $\partial S_1 / \partial \delta \alpha$, it is evident that the measurement S_1 contains additional information about the attitude beyond what is contained in S_0 . It is now desired to quantify how much additional information is contained in the S_1 measurement.

The local observability from measurement k is evaluated through the matrix $H_k^T H_k$, where H_k is the matrix of partials of the measurements with respect to the attitude:

$$H_k = \begin{bmatrix} \frac{\partial S_0}{\partial \delta \alpha_1} & \frac{\partial S_0}{\partial \delta \alpha_2} & \frac{\partial S_0}{\partial \delta \alpha_3} \\ \frac{\partial S_1}{\partial \delta \alpha_1} & \frac{\partial S_1}{\partial \delta \alpha_2} & \frac{\partial S_1}{\partial \delta \alpha_3} \end{bmatrix} \quad (75)$$

This observability measure is similar to the information contributed by that measurement, which is given by $H_k^T R_k^{-1} H_k$, where R_k is the measurement noise covariance and is assumed known for a given sensor. The trace of $H_k^T R_k^{-1} H_k$ allows the information that is contained about the observable direction of the system to be quantified. This is equivalent to the uncertainty measure used in [5], which uses $(1/\sigma_k^2)(\partial m_{\text{app}} / \partial \delta \alpha)^2$. This reference has m_{app} as the only measurement; therefore it is equivalent to the trace, where σ_k is the standard deviation of the measurement noise. Note that this reference takes the measurements to be the apparent magnitude, as opposed to the intensity S_0 . This results in an apparent increase in information as edge-on reflections are approached in the unpolarized case. In actuality, this behavior is due to the assumption that noise in the apparent magnitude, which is defined on a logarithmic scale, is constant. In actuality, this is not the case, but the noise in S_0 will be.

Figure 7 shows the ratio of information contributed by the measurement $\tilde{\mathbf{y}} = [S_0 \ S_1]^T$ compared to the measurement $\tilde{\mathbf{y}} = S_0$ for a single facet as a function of attitude. Figure 7a shows the gain for the case of zero atmospheric noise in S_0 or S_1 , e.g., a sensor in space. In this case, it is seen that there is a loss of information in every case, which means that the additional information contributed by the

polarization measurement is insufficient to overcome the loss of information due to the added noise of spreading the light out over the greater number of pixels needed to make the polarized measurement. Figure 7b shows the gain for the case of the magnitude and polarization calibration uncertainties of Sec. III.B. It is seen that for most attitudes, there is a significant information gain when polarized measurements are included. This information gain is maximum at edge-on reflection, which is where the reflection has the greatest polarization. A slight loss of information is still observed around a 45 deg reflection angle, which is expected because at this angle the reflection is unpolarized. Because the atmospheric contribution to uncertainty in polarization is less than to that in overall intensity, as larger aperture sensors are used, this information gain will become even larger.

V. Estimation of Attitude and Surface Material from Polarized Light Curves

Now that the observability of attitude and surface material have been established, a filter is run to demonstrate and assess the performance.

A. Simulation Parameters

Figure 8 shows the shape and material model that is employed in these simulations. The front of the solar panels, shown in blue, is modeled as gallium arsenide with 0.1-mm-thick BK7 cover glass. The “gold” faces are assumed to be multilayer insulation consisting of an 0.5 mm Kapton layer with an aluminum backing. The two “silver” sides on which the solar panel mount, the dish antenna, and back sides of the solar panels are assumed to be aluminum.

In this model, it is assumed that the BRDF surface roughness parameters n_u and n_v are equal and represented by $n_r \equiv n_u = n_v$. Here, it is assumed that BK7 glass has $n_r = 1000$, Kapton has $n_r = 515$, and aluminum has $n_r = 100$. An initial angular rate of $\omega = [-0.7 \ 0.3 \ 0.5]$ deg/s is used to initialize the simulation.

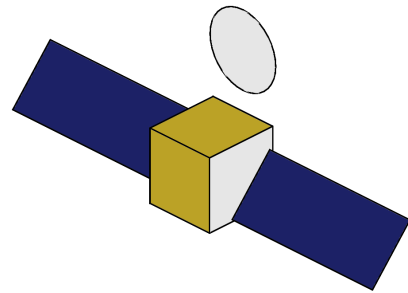


Fig. 8 Multimaterial box-wing model.

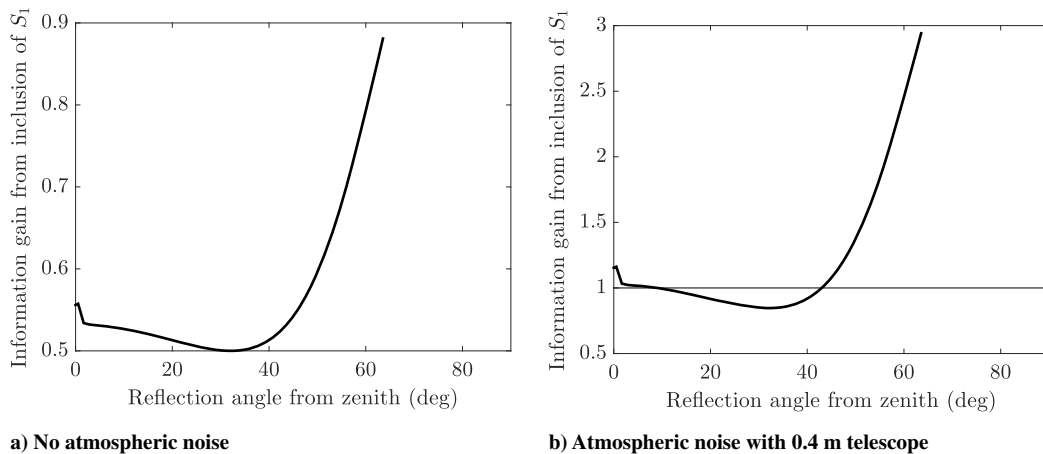


Fig. 7 Attitude information ratio from inclusion of S_1 parameter.

B. Simultaneous Attitude and Surface Material Estimation using MMAE

Now, MMAE is used to estimate the attitude, angular rate, surface roughness coefficient n_r for each material, size scale-factor s , and surface materials of the spacecraft. The observability of n_r has been established in [5]. In this case, the estimated states are

$$\mathbf{x} = [\delta \mathbf{p} \ \omega_{B/I}^B \ n_r \ s]^T \quad (76)$$

where $\delta \mathbf{p}$ is the local-error GRP in the UKF and s is a scale factor for the model. MMAE is used to estimate the materials, with different models in the model bank having different material compositions. Process noise is constant across all filters in the bank and has been tuned to account for disturbance torques. Table 1 shows the specular material bank used in the MMAE models, and Table 2 shows the diffuse material bank. In this simulation, 500 different material composition models are randomly generated from the materials in Tables 1 and 2. Each surface is assigned a discrete material. All weights in the MMAE process are set equal at the beginning of the filter. First, results will be shown assuming that the true model is contained within the model bank, then results will be shown if the true model is not contained. In these simulations, the UKF is initialized with an initial attitude error of 5 deg and initial rate error of 0.5 deg per second. Initial attitude and rate uncertainties (1σ) are assumed to be 15 deg and 0.5 deg per second, respectively. An initial scale factor error 1.5 is used, with an initial scale factor uncertainty (1σ) of 2. The measurement noise statistics are $\sigma_{S_0} = 3.1 \times 10^{-16}$ W/m² and $\sigma_{S_1} = 6.6 \times 10^{-17}$ W/m², which are typical for a CCD sensor on a 0.4 m telescope.

Table 1 Specular material bank

Material	Refraction index n	Extinction coefficient k
Aluminum	1.1987	7.0488
Gold	0.2773	2.9278
Copper	0.6366	2.7834
Nickel	1.9648	3.8352
Tin	2.1600	6.3500
Titanium	2.9300	3.9850
Zinc	1.5020	6.5574
Iron	2.8735	3.3590
Cobalt	0.5690	6.8015
Silver	0.2893	5.3184
Platinum	0.5189	7.2743
Lead	1.8800	3.4900
Tungsten	3.8868	2.7458

Table 2 Diffuse material bank

Material	Refraction index n	Extinction coefficient k	Diffuse reflectance
Gallium arsenide	3.857	0.198	0.02
Crystalline silicon	3.4	0.2	0.03
Polymide ("Kapton")	1.746	9.273×10^{-5}	0.300
BoPET ("Mylar")	1.636	0.00	0.65
PTFE ("Teflon")	1.31	0.00	0.98
Silicon carbide	2.26	6.35	0.21
Thallium bromide	2.352	0.00	0.163
"Berkeley Black"	2.00	7.00	0.025
Titanium dioxide pigment	2.613	0.00	0.82
Synthesized black coating 1	2.20	1.00	0.04
Synthesized black coating 2	2.30	1.50	0.04
Synthesized black coating 3	2.40	2.00	0.03
Synthesized black coating 4	2.50	2.50	0.03
Synthesized white coating 1	1.50	0.00	0.90
Synthesized white coating 2	1.60	0.00	0.90
Synthesized white coating 3	1.70	0.00	0.90
Synthesized white coating 4	1.80	0.00	0.90

Figure 9 shows the attitude and angular rate errors for this case. Note that in Fig. 9 and all subsequent filters showing estimation performance results, the blue line represents the errors, while the red lines represent the $\pm 3\sigma$ bounds. These labels are omitted from each figure for clarity of the figure. Figure 10 shows the surface roughness errors. The large drops in the errors and covariances for a particular material occur when that material glints, which is richer in information than diffuse reflections. Figure 11 shows the convergence of the models containing the various materials as well as convergence of the size scale-factor parameter. From these results, it is seen that in the presence of the correct model in the model bank, the correct surface material model is identified, and the attitude, angular rate, surface roughness, and size scale-factor estimates all converge.

In real applications featuring complex models, however, it is unlikely that a perfect model match will exist in the model bank. Therefore, it is desired to evaluate expected performance in this case. To do so, the correct model, as well as very similar models (e.g., those that are identical except for one surface) are manually removed from the model bank. It is ensured that at least three surfaces are mismatched between any model in the bank and the truth model. Figure 12 shows the MMAE weights and the size scale-factor estimate for this case. The model that is identified has incorrect materials for the solar panels and the sides of the bus to which the solar panels are attached. It is seen that the size scale-factor converges, but not to zero error. This is due to the modeling error; the size scale-factor is compensating for the differences between the true model and the assumed model. In this case, the size is being underestimated by approximately 0.6%. The solar panel material is identified as crystalline silicon and the bus faces to which they are attached are identified as zinc. Crystalline silicon is another material from which solar cells are made, and it has similar optical properties to gallium arsenide, which is the true solar panel material. Zinc has similar optical properties to aluminum, which is the true material for the bus sides. Figure 13 shows the convergence of the attitude and angular rate errors in this case. The assumed measurement noise in the filter is manually increased until convergence of the estimates is observed. It is found through this tuning that when the measurement noise is increased by approximately a factor of four, the attitude estimates become convergent. Monte Carlo runs are shown in Fig. 14, which show that although errors sometimes fall outside of 3σ due to the large errors in the assumed model, the estimates still converge toward the true attitude, and attitude errors reliably converge to within 1 deg. It is again noted that there are large errors in the assumed filter models, which cause the errors, although convergent, to be often poorly bounded by 3σ . If the model bank were increased in diversity such that a better match were located in the model bank, this performance would improve.

From this analysis, it is evident that MMAE is a plausible method to perform simultaneous attitude and surface material estimation of an RSO from polarized light curves. Although increasing complexity in the shape model will result in a larger number of models required in the filter bank, it is found that models that are a poor match have their weights driven to zero quickly, thereby reducing the computational overhead of running MMAE with a large model bank. Additionally, the method is found to be tolerant to lack of a perfect model match in the model bank. It is found that if there is a similar model in the bank, then MMAE will choose that model, and the attitude estimates converge accurately. The size scale-factor estimates are found to converge but may overestimate or underestimate the true size. However, the error in the size estimates is found to be small. Therefore, it is concluded that MMAE can be robust to tolerating modeling errors, and allow for simultaneous attitude, size, and surface material estimation.

C. Impact of Depolarization

As discussed previously, the polarized BRDF model developed in this work assumes only a single reflection from each facet. In reality, there may be some multiple reflections, where light reflects off one facet onto another, and from there either to the observer or onto yet another facet. This case is depolarizing. The lack of empirical models

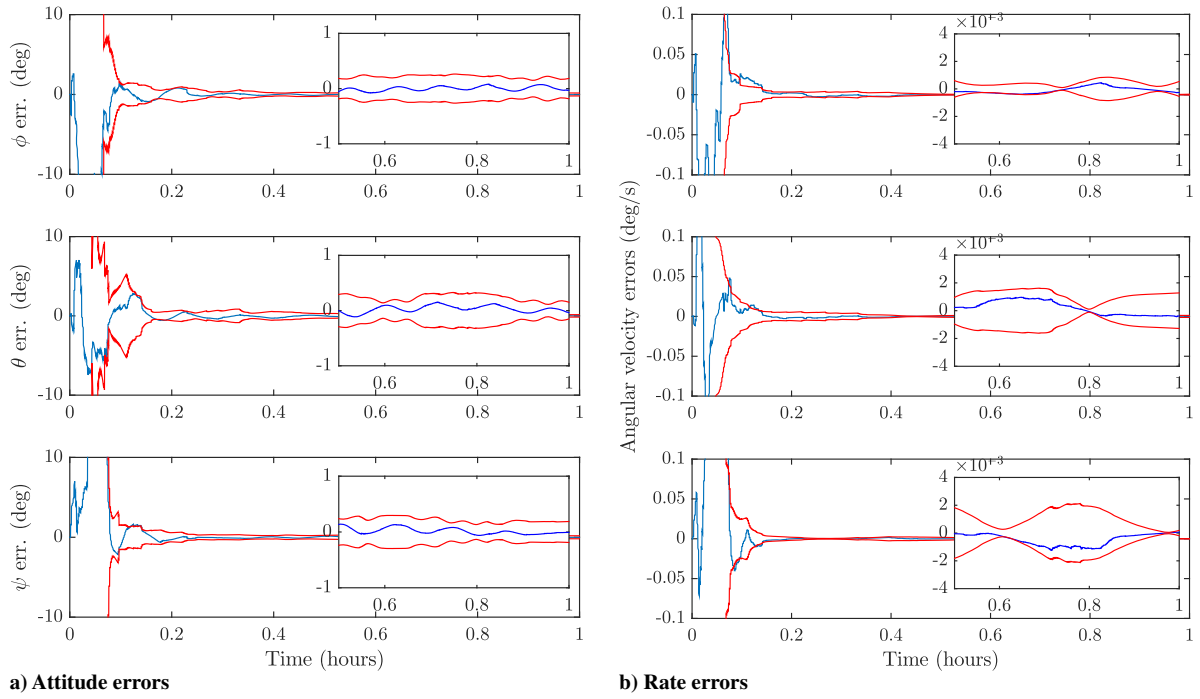


Fig. 9 Convergence of surface roughness parameters for multimaterial case.

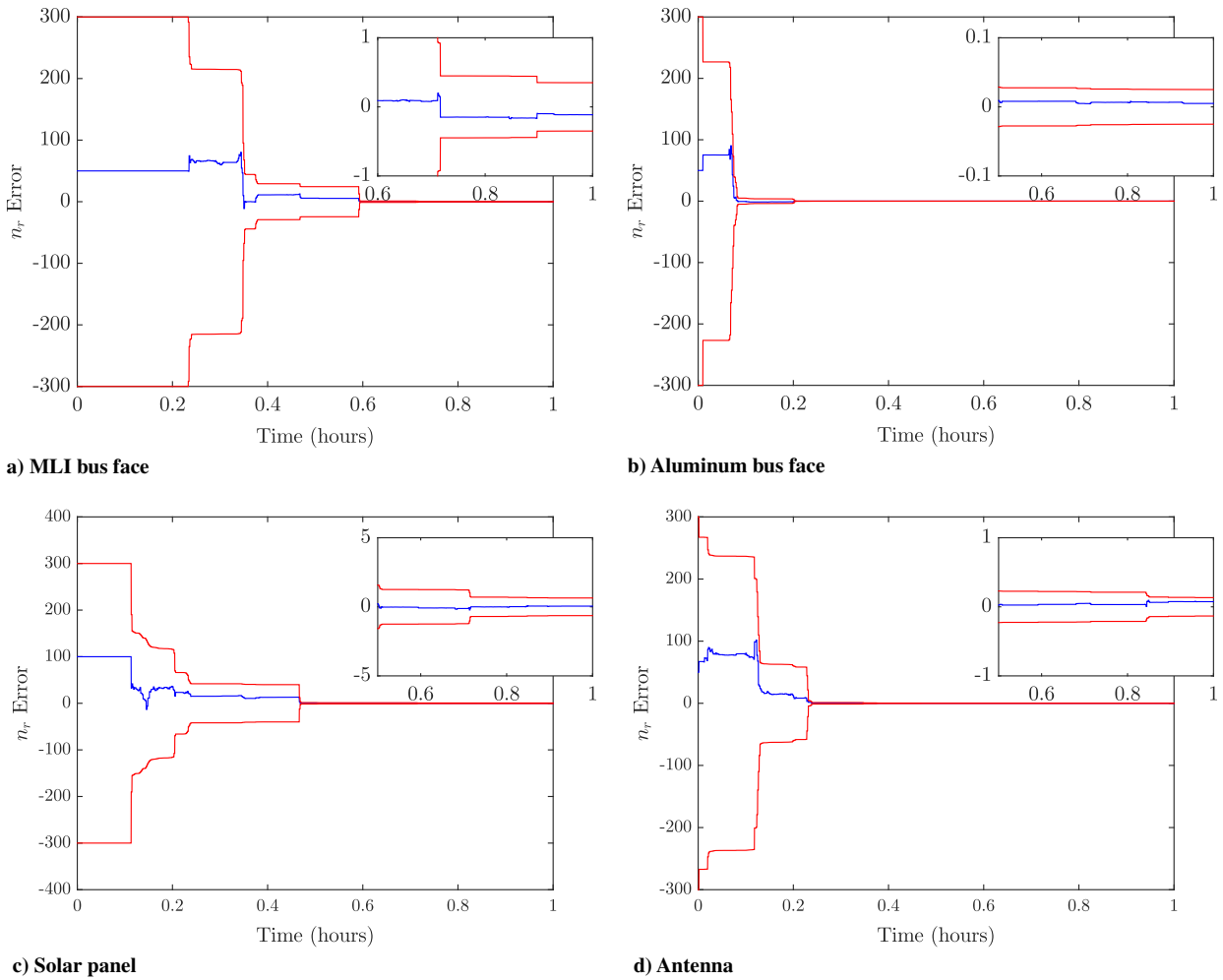


Fig. 10 Material model and surface parameter estimation results.

to address this has been previously discussed. Therefore, it is desired to assess the impact of unaccounted depolarization on the estimator accuracy. To assess this, the filter is now run with a varying level of

depolarization modeled in the S_1 parameter. It is noted that, in this case, the model bank contains a perfect match. The depolarization is assumed to be constant, in the absence of a model that relates it to

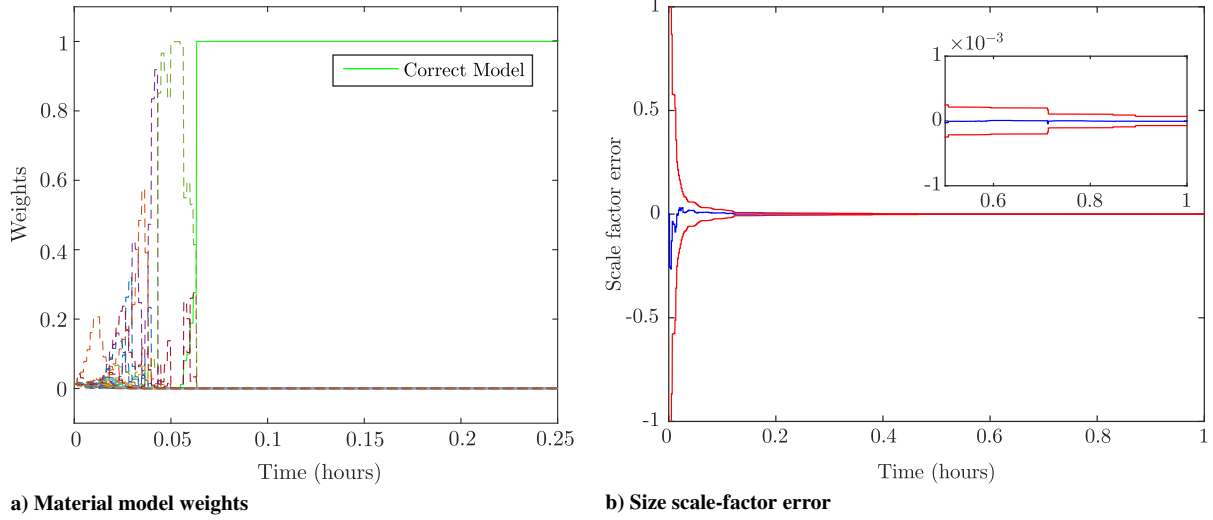


Fig. 11 MMAE weights and size scale-factor convergence with large model mismatch.

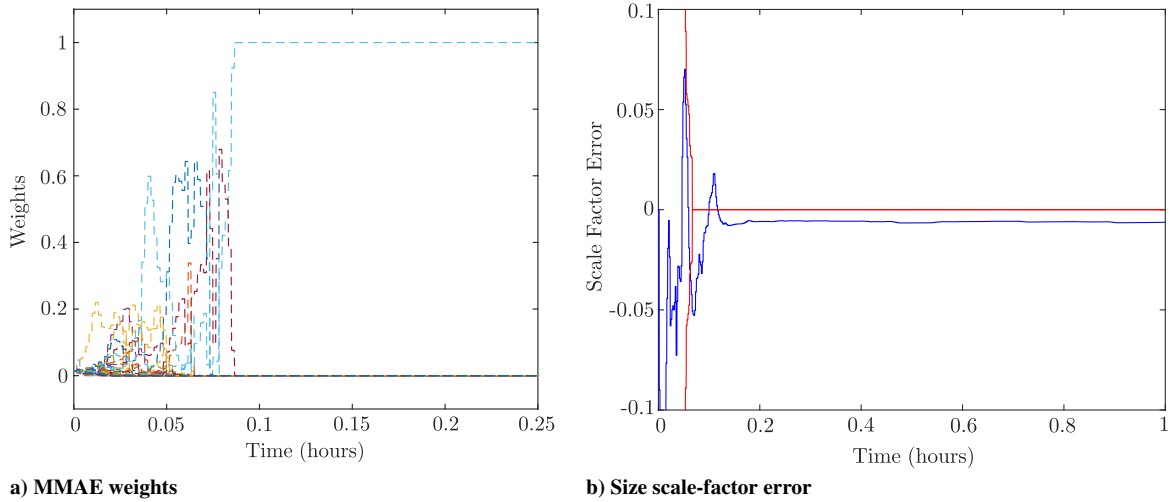


Fig. 12 Attitude estimates for MMAE with large multimaterial model bank with large modeling error.

surface properties and orientation. This is unlikely to be a limiting assumption, as it creates an unknown bias in the measurement as opposed to a noise term. To assess the impact of depolarization, the ratio of errors between the polarized and unpolarized filters is assessed for varying levels of depolarization. This shows whether the inclusion of mis modeled polarization information is detrimental or beneficial to filter performance compared to the unpolarized case. Figure 15 shows the ratio of the total attitude estimation error of the polarized filter to the unpolarized filter. It is seen that, for 1% or less depolarization, the polarized filter still performs better. At 2% depolarization, the error average is approximately equal. Although at some times the polarized filter will perform better, at others the unpolarized will perform better. At 3% depolarization, the unpolarized filter performs better, on average. Thus, it is concluded that the benefit of polarized measurements to attitude determination performance depends on being able to know the depolarizing contributions of a surface. This underscores the need to develop better models for determining the depolarization of reflection off a diffuse surface. Figure 16 shows the attitude estimate errors and 3σ bounds from the filter covariance matrix for both the polarized and unpolarized filters assuming 3% depolarization. The depolarization has the effect of increasing the errors beyond the expected 3σ bounds.

In real applications, the true attitude state will be unknown, and thus the true error cannot be computed. However, it is still desirable to be able to assess whether the inclusion of the polarization measurement aids or hurts the filter performance. Although true state data are unavailable, analysis of the measurement residuals can give insight

into whether the dynamics and measurement models accurately describe the system. The Mahalanobis distance is a measure of the distance between a point and a distribution, and it can be used to quantify how well the measurement residuals fit within the expected measurement covariance from the filter. The Mahalanobis distance assumes that the underlying distribution is Gaussian. In this case of a perfect material model match, the errors and measurement residuals pass a Anderson–Darling test at the 5% significance level, which indicates that the distribution is largely Gaussian and that Mahalanobis distance is an appropriate measure. It is noted that, for the case of a large model discrepancy, as in Fig. 14, the errors do not pass the Anderson–Darling test at the 5% significance level, and as such caution should be exercised if using the Mahalanobis distance to assess measurement residuals for that case. This suggests that having a larger model bank to contain a model closer to the truth is of increased importance if significant depolarization exists.

For this case, the Mahalanobis distance for measurement k can be written as

$$d_k = \sqrt{(\tilde{\mathbf{y}}_k - \hat{\mathbf{y}}_k)^T (P_k^{vv})^{-1} (\tilde{\mathbf{y}}_k - \hat{\mathbf{y}}_k)} \quad (77)$$

where $\hat{\mathbf{y}}_k$ is the output estimate, and P_k^{vv} is the residual covariance given by the UKF. It is desired to determine the probability that the measurement lies within the covariance distribution calculated by the filter. The threshold to compare the Mahalanobis distance to is obtained from the inverse χ^2 cumulative distribution at a significance

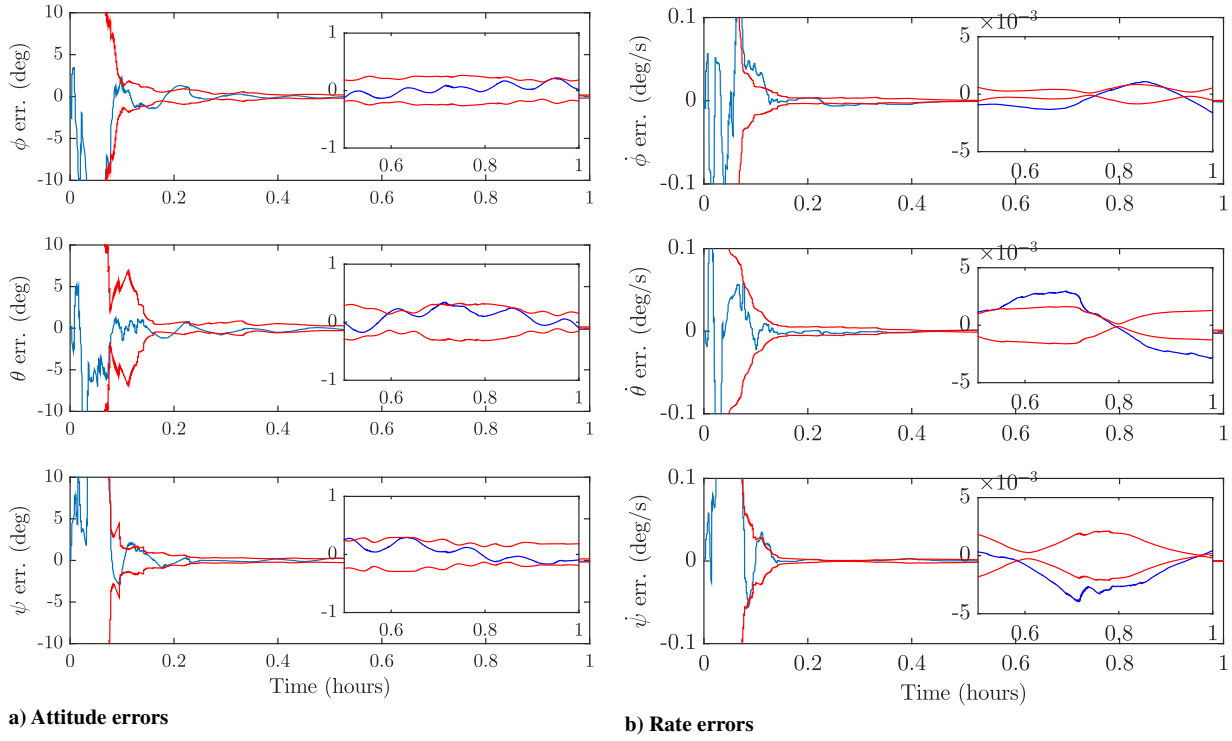


Fig. 13 Monte Carlo attitude estimation runs for MMAE with large multimaterial model bank with large modeling error.

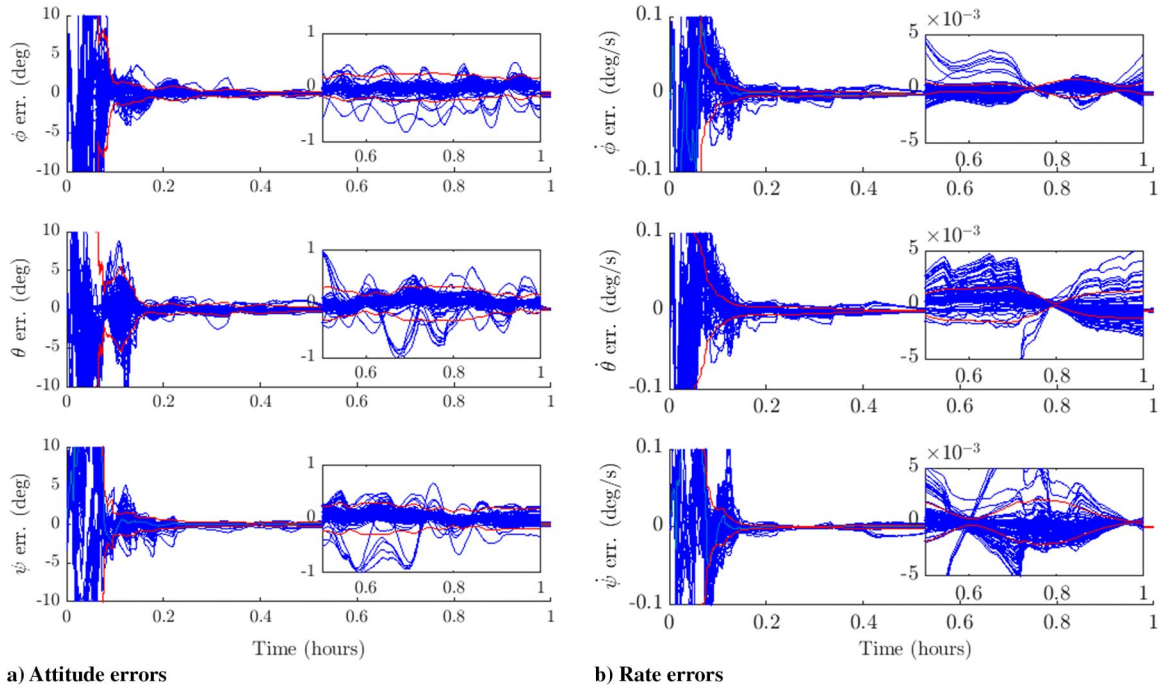


Fig. 14 Error ratio between polarized and unpolarized filter for different levels of depolarization.

level α . The solid ellipsoid of values satisfying $d_k^2 \leq \chi_n^2(\alpha)$ has probability $1 - \alpha$. For 99% confidence or $\alpha = 0.01$, with three degrees of freedom, $\chi^2 = 11.34$. Figure 17 shows the Mahalanobis distances for measurements with different levels of depolarization plotted alongside this χ^2 threshold. Here, multiple instances of the filter are run, each with a measurement set that has been depolarized by a constant amount. The Mahalanobis distance of the measurement residuals for each of these filters is overlaid into the plot of Fig. 17, with dots corresponding to each residual, colored by which measurement set they were drawn from. It is seen that, for the unpolarized measurements as well as the polarized measurements with 0%

depolarization, the Mahalanobis distance remains well under the χ^2 threshold. As the depolarization increases, the Mahalanobis distance more frequently exceeds this threshold. For 1% depolarization, only occasional measurements lie outside the threshold. At between 3 and 4% depolarization, a significant amount of the measurements lie outside the threshold. This corresponds well to the levels of depolarization found to degrade filter performance seen in Fig. 16. As such, it is seen that a χ^2 test on the measurement residuals can be used to assess whether the polarization is being modeled accurately enough and whether the polarization measurements should be used in the filter. However, it is again noted that these results are valid

under the assumption sufficiently correct model is available to give Gaussian errors.

D. Tolerance to Initial Attitude Errors

One of the shortcomings of attitude estimation from light curves, using the methods shown in [8,10,28], is the need for a good initial estimate to ensure filter convergence. The unpolarized filter has been found to be unable to handle a 2 deg/s error in the initial rate estimate. This has been one of the challenges in implementing this type of algorithm in real-world applications. This is primarily driven by the nonuniqueness of solutions that may locally minimize the error residual in a nonlinear system. The problem of determining attitude from light curves is plagued by this issue. Assuming that the RSO has multiple reflecting facets and there are three or more observers on the ground, the static attitude determination problem from light curves is fully observable. However, if least-squares is used to estimate the static attitude (at a single instant), the initial guess must be close to the truth for the correct solution to be obtained. Otherwise, there are typically many attitudes that locally minimize the measurement residual. In the dynamic case, each local minima will evolve differently with time. While this will typically prevent the filter from converging on the wrong minimum, if the filter initially converges toward the wrong minimum, it is prone to bounce between minima and not converge on the correct solution. An example of this situation is illustrated in Fig. 18.

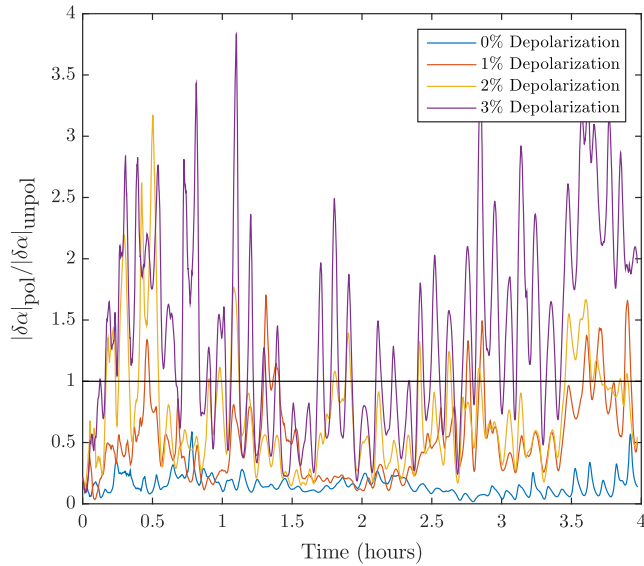
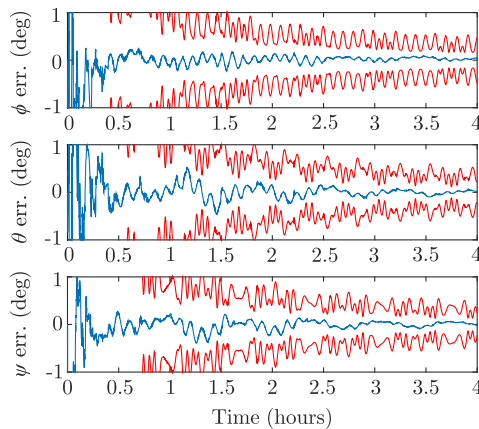
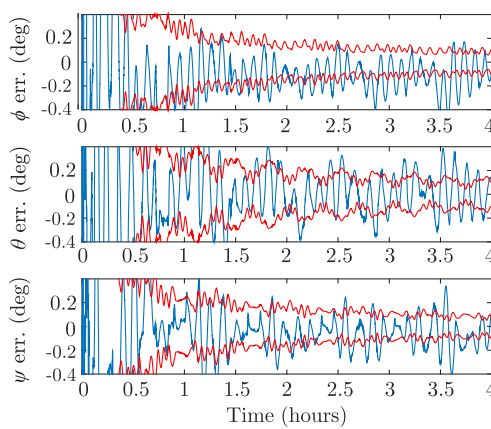


Fig. 15 Estimate errors and 3σ bounds for unpolarized and polarized measurements, assuming 3% uncertainty and sensor case 3.



a) Unpolarized



b) Polarized

Fig. 16 Mahalanobis distance of measurement residuals and χ^2 threshold.

Let $\Delta \mathbf{y}$ be the measurement residual, which is computed as $\Delta \mathbf{y} = \tilde{\mathbf{y}} - \hat{\mathbf{y}}$, where $\hat{\mathbf{y}}$ is the expected output from the state estimate. Then, the least-squares cost function J is [29]

$$J = \Delta \mathbf{y}^T \mathbf{R}^{-1} \Delta \mathbf{y} \quad (78)$$

Minimizing J minimizes the measurement residuals $\Delta \mathbf{y}$, weighted by the measurement noise covariance $\mathbf{R}$. Figure 18 shows the values of the cost function across a two-dimensional attitude profile defined by θ and ψ (where the first rotation ϕ is constant). In the first case, only the unpolarized measurement is assumed, with $\mathbf{y} = \mathbf{S}_0$. In the second case, polarized measurements are assumed, with $\mathbf{y} = [\mathbf{S}_0 \ \mathbf{S}_1]^T$. In the unpolarized case, the cost function is minimized in a continuous “ring.” This indicates that there is some axis about which a rotation ϕ will not cause a change in the reflection intensity toward the observer. When the polarization measurement is added, this ring becomes a set of four distinct local minima in this area. Therefore, the polarization helps to resolve some of the ambiguity in the rotation ϕ . While the filter can still be driven toward the wrong minimum, which potentially could cause it to diverge, the addition of polarization information makes it more likely that the filter will be driven to the correct solution. This allows for greater tolerance of initial errors.

The filters are now run with varying initial errors to assess their tolerance. Table 3 shows the maximum initial attitude and rate errors that result in filter convergence for filters using polarized and unpolarized measurements with varying degrees of depolarization. It is

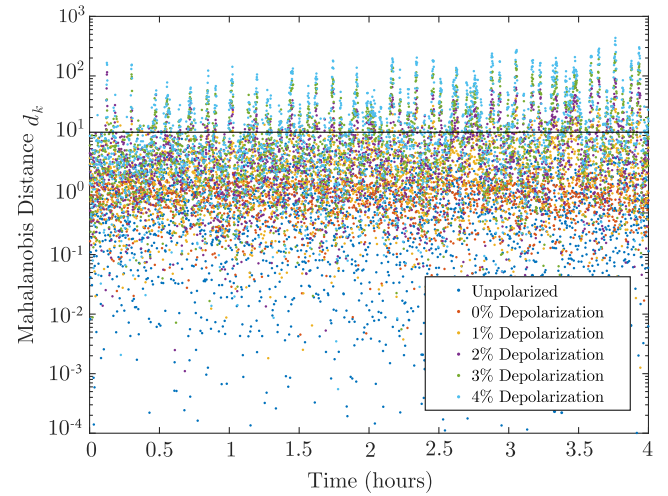


Fig. 17 Local minima of least-squares cost function for unpolarized and polarized measurements.

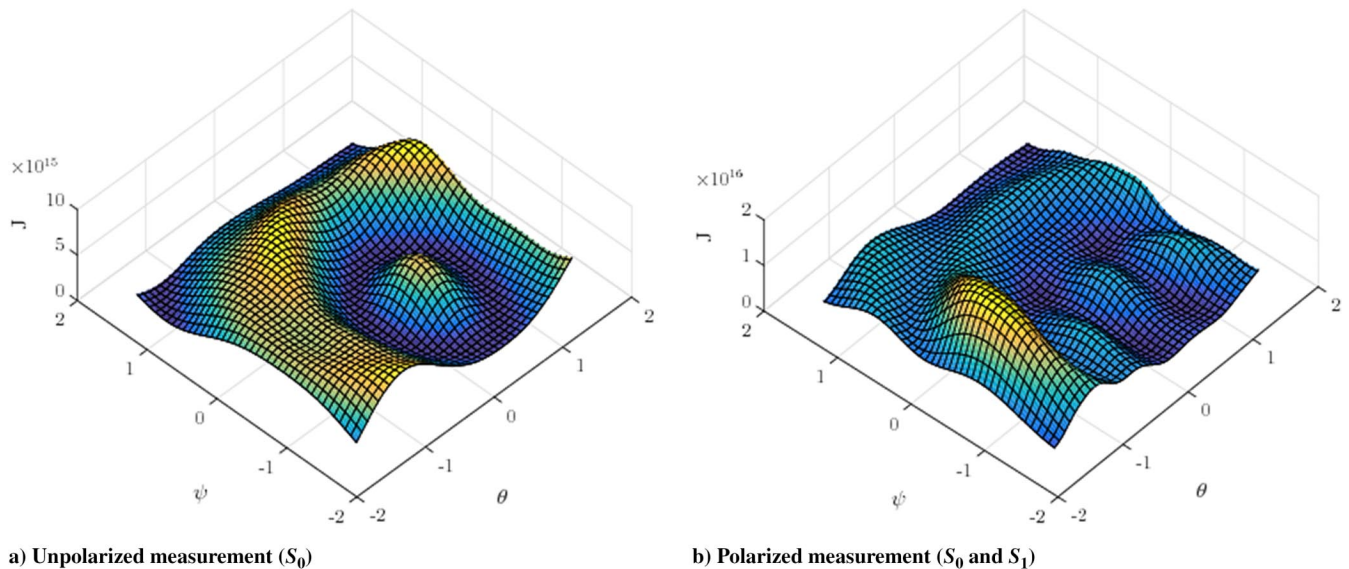


Fig. 18 Local minima of least-squares cost function for unpolarized and polarized measurements.

Table 3 Maximum initial errors with depolarization				
Rate error (deg/s)	Attitude error (deg) unpolarized	Attitude error (deg) 0% depolarization	Attitude error (deg) 3% depolarization	Attitude error (deg) 4% depolarization
0	54	58	58	55
0.5	42	51	46	42
1	18	37	29	2
2	N/A	0.6	N/A	N/A

seen that the addition of polarized measurements allows for larger initial errors. Unmodeled depolarization, however, decreases the allowable initial errors. Unmodeled depolarization of up to 3% still results in larger initial error tolerance than just unpolarized measurements, which is about the same amount of depolarization that gives added benefit in the attitude filter.

VI. Conclusions

In this work, the use of polarized light curves for SSA was investigated. A polarized BRDF was developed based on Ashkmin and Shirley’s unpolarized BRDF function. The developed model considers both the polarization of specular and diffuse reflections. The polarization of the specular reflection is found to depend on the object’s surface material but not attitude, while the polarization of the diffuse reflection depends on both surface material and attitude. The observability of attitude and surface material parameters from polarized light curves is assessed. It is determined that polarization measurements contain information about the surface material, but direct observability of the refraction and extinction coefficients does not exist. However, surface materials are able to be estimated using multiple-model adaptive estimation. It is also determined that, for sensors in the atmosphere, the lesser atmospheric effects on the polarization than the overall intensity lead to increased attitude observability for most attitudes when polarized sensors are used. A filter is implemented to estimate the object’s attitude, angular rate method is robust even if perfect material model matches do not exist in the model bank. It is also found that the addition of polarized measurements allows for larger initial errors in the filters to be tolerated.

The modeled polarization of diffuse reflections highly depends on the assumption that all diffuse reflections occur due to subsurface scattering, and that a light ray does not reflect off multiple microfacets. While this is expected to hold mostly true for artificial objects, little prior work has been performed to assess the impact of surface finish on depolarization of the light curve. The impact of this unmod-

eled depolarization has been considered. It is found that attitude estimation performance is somewhat sensitive to this depolarization; after approximately 3% depolarization, the attitude estimation performance becomes worse than it would be if polarization information is omitted. A χ^2 test on the Mahalanobis distance between the expected measurements computed from the filter and actual measurements can be used to determine if this phenomenon is occurring, and to determine whether the unpolarized or polarized measurements should be used.

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